MATHEMATICAL CLOSURE REVIEW · DOCKET T1–T9

Mathematical Closure Toward a General Theory of Sustainability

Audit of the canonical schema TCS-1.0; theorem dependency graph for the open docket; verdicts and complete result records for the nine targets T1–T9, with self-contained proofs and counterexamples; the universal / model-class boundary theorem; and a dependency-ordered research plan.

Z.ai — External Mathematical Review

Prepared against the self-contained General-Theory Mathematical Closure Packet (TCS-1.0; immutable A001–A025 sources; controlling corrected-theorem records; result records R01–R09 in the required output schema).

Contents

> **Typesetting note (PDF rendering only).** The canonical packet typefaces for schematic objects (double-struck K, W, R, M, T, P; fraktur A, S, B; sans-serif Z, I, P, D, O, U, W; script V, D, A, K, S, Y) are rendered in this PDF as the corresponding plain letters (K, W, R, …). The markdown source files retain the canonical forms and remain the documents of record. Long arrows are rendered as their short equivalents.

Master Mathematical Closure Review — General Theory of Sustainability

Packet: `general_theory_math_closure_packet` (self-contained, per `14_SELF_CONTAINMENT_REPORT.md`) **Controlling objects:** `TCS-1.0` (`control/01_canonical_system_schema_TCS_1_0.md`), corrected theorem records (`corrected_theorems/01–09`), immutable A001/A002 sources (`sources/`), docket `11_OPEN_THEOREM_DOCKET.md` (targets T1–T9), output schema `12_REQUIRED_OUTPUT_SCHEMA.md`. **Reviewer role:** per `10_MASTER_REVIEW_PROMPT.md` (senior mathematician spanning viability theory, differential inclusions, hybrid/RFDE systems, robust control, set-valued analysis, aggregation, stochastic viability, compositional systems). **Result-record files:** `01_result_records/R01–R09`, one per docket target, each in the required 17-field schema. This document carries: the schema audit (§1), the theorem dependency graph (§2), the verdict table (§3), the universal/model-class separation (§4), the source-unity vs. publication-dependence audit (§5), the minimum new-theorem set (§6), and the dependency-ordered research plan (§7). **Status discipline:** every claim below carries either a packet-internal proof/anchor (file + label/line), a complete proof in a result record, or an explicit open/conditional flag. No external bibliographic verification was available in the packet; all novelty fields are correspondingly qualified (`14_SELF_CONTAINMENT_REPORT.md`: “novelty adjudication and exact bibliographic theorem-number verification still require the cited publications”).

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1. Audit of the canonical schema TCS-1.0 (docket task 1)

The audit examines `control/01_canonical_system_schema_TCS_1_0.md` (with its JSON freeze) for missing types, incompatible quantifiers, and hidden circularity. Findings are graded: **[GAP]** missing type/field, **[QF]** quantifier conflict needing a guard, **[CIRC]** circularity risk needing an explicit acyclicity declaration. Each finding ends with the minimal repair; none requires abandoning the frozen schema — all are additive and belong in a `TCS-1.1` migration as foreseen by §10 of the schema.

1.1 Missing types

[GAP-1] Generation/specification-change index is not a typed object. `TCS-1.0` §1 states that changing a threshold, authority, population, horizon, information pattern, or action class changes `Ω`, but the schema has no type for a declared sequence `(Ω_g)_{g∈G}` of specifications with transition maps between the corresponding admissible sets and architectures. Target T7 (intergenerational continuation) cannot even be stated inside the current type system without smuggling the generation structure through an ad-hoc construction. A001 §14 (Definition 14.1, `sources/A001_topdown_source.txt` line 1752) already uses generation-indexed constraint sets `V^{(k)}` — this exists in a source but not in the canonical types. **Repair:** add a `specification_path` type: a locally finite index set `G`, a map `g ↦ Ω_g`, and per-transition typed translation maps (state, identity/liability/obligation, harm). Record R07 instantiates exactly this and can serve as the migration template.

[GAP-2] Forward-completeness / confinement certificate is not a typed field. The theorem-record schema §6 (Θ) lists `premises` and `horizon` but has no dedicated field for the forward-completeness or compact-confinement certificate. Yet the controlling corrected theorems demand it: the strong-invariance theorem (corrected `02`, assumption 2), the restricted composition theorem (corrected `03`, assumption 4), and the finite/infinite-horizon non-implication (corrected `01`, non-implication 8: "Finite-horizon viability does not imply infinite-horizon viability without a closure/compactness argument"). The compactness lemma proved in record R03.2 shows the field is load-bearing, not cosmetic. **Repair:** add `confinement_certificate` to Θ with values {`forward_complete`, `compact_enclosure(K̂)`, `none`} and require it non-`none` for any infinite-horizon claim.

[GAP-3] Erosion/error budget triple is not a first-class typed object. Mapping type 5 (`APPROXIMATION`) of §7 requires "error, domain, horizon, and safety erosion" to be explicit, but the Θ record has no structured field for the erosion triple `(ε, r, α)` that the conditional tubular erosion lemma (corrected `02`, Lemma 2: `L_G r + Δ_ε ≤ α`) makes canonical. The required output schema (field 11) asks for it per result; the canonical record should ask for it per mapping. **Repair:** add an `erosion_triple` sub-record to every `APPROXIMATION` concordance row: `(error_budget Δ_ε, erosion depth r, boundary margin α, Lipschitz gain L_G, tubular radius ρ)`, with the feasibility condition `L_G r + Δ_ε ≤ α` recorded as a checkable contract. Records R03 and R05 both populate this structure.

[GAP-4] Observation kernel typing (deterministic set-valued vs. stochastic) is under-specified. §2.3 declares `O_q` as "observation map/kernel" without separating (i) deterministic set-valued observations `Y = O(x)` (used by the epistemic-viability chain, corrected `01` §Epistemic viability via `Compat(b)`), (ii) stochastic observation kernels (required by chance viability, §4.6, which needs an "explicit law and filtration"). The two are different types with different fibre semantics; conflating them is precisely the type error the schema elsewhere prohibits. **Repair:** type `O_q` as a tagged union {`deterministic_setvalued | stochastic_kernel(filtration-aligned)`}; require the chance-viability judgment to declare the stochastic branch with its filtration and the support condition of Proposition 8 (corrected `01`).

[GAP-5] Cumulative-harm/obligation block has no declared transition semantics at architecture resets. §2.1 lists `h` (cumulative harm, obligation, accounting states) as a state block, and Operator II (§5) requires "cumulative identity/liability/harm constraints" in the transition-safe sets — but no typed rule says how `h` translates across an architecture reset (additive accrual? capped carry-over? forgiveness event?). R07 (§2, obligation translation) shows the choice is mathematically decisive: monotone accrual with a harm floor yields a finite support horizon (R07 Corollary 3), whereas forgiveness resets change the judgment entirely. **Repair:** declare h-translation as part of the Operator II reset data R_{qe}, with named variants {accumulate, cap, forgive}.

1.2 Incompatible quantifiers needing guards

[QF-1] The canonical judgments in §4 do not bind the solution concept per judgment. §4 defines the eight judgments "for fixed (Ω,q,T) and aligned classes" — the solution concept S appears only in the hierarchy document (corrected 01, signature Ξ). Since the same Ω,q,T can admit ODE, Filippov, and RFDE solution concepts with different solution sets, two records could disagree while claiming the same judgment. The hierarchy's guard ("Empty solution sets never make a safety statement true") is controlling but lives outside the frozen schema. **Repair (TCS-1.1):** bind S into every judgment record; import the empty-solution guard as an axiom-level clause of causal admissibility.

[QF-2] Chance viability lacks a support-alignment clause in the schema. §4.6 defines chance viability "under an explicit law and filtration" but does not require the declared law's support to be related to the robust disturbance class. Proposition 8 (corrected 01) is exactly the missing guard: robust-to-chance implication holds only "if the stochastic law is supported on the robust disturbance class", and "no implication is available when the stochastic support exceeds the robust class". **Repair:** add to §4.6: "A chance-viability record must declare supp(law) ⊆ W or supp(law) ⊄ W; only the former may cite Proposition 8." Counterexample (c) in record R08 instantiates the failure.

[QF-3] Implementation-branch quantifier is defaultable and therefore dangerous. §2.3 says prescribed and realized actions are different types; the hierarchy (Proposition 7) proves that existential (some branch) and universal (every branch) implementation semantics are not mutually monotone: enlarging the implementation correspondence shrinks all-branches safety and enlarges some-branch safety. The schema permits an institution record without declaring which semantics it uses. **Repair:** make the branch quantifier ∀u^real ∈ I_q vs ∃u^real ∈ I_q a mandatory declared field of institutional viability; forbid silent switching (the flip is an axiom-4 violation when it rescues a claim).

[QF-4] Recoverability's three quantifier levels (robust/existential/disturbance-strategic) need explicit declaration. §4.7 defines recoverability "robustly or existentially according to the recorded quantifiers" — acceptable as a schema, but the docket T2 acceptance and the corrected A001 audit (corrected 06 §7: Theorem 5.2's repair turns "every trajectory exits" into an adversarial-selection statement) show the mis-recorded version has already occurred in sources. **Repair:** require the disturbance quantifier pattern (pure ∀w, adversarial nonanticipative strategy, or ∃w) as a named field.

1.3 Hidden circularity checks

[CIRC-1] Admissible-set generation from the typed registry — benign if and only if the registry is layered. K_{q,Ω}⊂Z_q is "generated from the typed registry" including epistemic conditions (§2.5). Since the information state b is itself a state block (§2.1), an epistemic constraint inside K constrains the estimate, not the plant — a physical viability judgment computed against such a K would silently reward optimistic filters (smaller compatible sets make epistemic constraints easier). No circularity arises if physical and epistemic constraint blocks are kept as separately typed generators of K; the danger is only in an unlayered registry. **Verdict: no circularity in the corrected hierarchy (physical/epistemic judgments live on different spaces with typed maps, corrected 01), but the schema must state the layering.** Repair: declare K_{q,Ω} = K^{phys} ∩ π^{-1}_{info}(K^{epi}) with the two generators separately typed, and forbid physical judgments from citing K^{epi} blocks.

[CIRC-2] Composition gate refers to an open class of theorems — bounded, not circular, but currently ill-typed. §5 admits composition "after … an applicable invariance/small-gain theorem" — a gate whose content depends on which theorems exist. This is status discipline, not circularity, but as written the gate is uncheckable because the admissible theorem list is not enumerated. **Repair:** enumerate the controlling records: (i) the corrected restricted proximal-normal composition theorem (`03`); (ii) new: the tubular assume–guarantee theorem of record R05 (this review); (iii) explicitly excluded until proved: nonlinear small-gain with nonconvex implementation, variable-event hybrid composition. Composition citing anything else is `UNRESOLVED`.

[CIRC-3] Implementation map must be declared non-strategically. `I_q` maps prescriptions to realized actions. If `I_q`'s declaration were allowed to depend on the policy class `P_q` (e.g., compliance responses tailored to the specific policy), then institutional viability `∃π ∀u^real ∈ I(π,·)` would be a fixed-point condition across the declaration boundary — a genuine circularity. §2.3 does not forbid it. **Repair:** require `I_q` declared before and independently of `P_q`; strategic compliance (I responding to the policy) must be re-typed as a game model class with equilibrium solution concept, outside the present judgment family. This matches the master prompt's non-negotiable rule "One causal policy must be distinguished from disturbance-observing controls" and extends it to the implementation side.

[CIRC-4] Operator II's "admits the proved … predecessor construction" — sound, keep. The gate references a specific proved record (corrected `04`), so no circularity; R01 adds the missing converse (what the theorem would silently certify if tubes were replaced by endpoints — nothing, provably).

[CIRC-5] Status monotonicity vs. theorem-record promotion — already guarded. Axiom 5 (status monotonicity) plus the concordance requirement blocks the classic circularity in which an integrated narrative promotes a conditional source claim to `proved` because a neighbouring corrected theorem "covers" it. The master prompt's rule "Do not strengthen a source claim by proximity to a corrected theorem" is enforced by the schema; no change needed.

1.4 Audit verdict

`TCS-1.0` is structurally sound: no destructive circularity, and the mapping vocabulary (§7) is adequate for every bridge examined in this review. The five gaps and four quantifier guards are all additive; the recommended `TCS-1.1` diff is: `specification_path` type (GAP-1), `confinement_certificate` field (GAP-2), `erosion_triple` field (GAP-3), tagged observation types (GAP-4), `h`-translation variants (GAP-5), per-judgment `S` binding (QF-1), chance-support clause (QF-2), mandatory implementation-branch quantifier (QF-3), recoverability quantifier pattern (QF-4), registry layering (CIRC-1), enumerated composition gate (CIRC-2), non-strategic `I` declaration (CIRC-3). Every result record below is written so as to be valid under both `TCS-1.0` and the proposed `TCS-1.1`; where a record uses a proposed new type (R07's specification path, R03/R05's erosion triples), this is stated in its field 16 (remaining obligations).

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2. Theorem dependency graph for the open docket (docket task 2)

Proved bases on which the docket targets rest (all packet-internal):

- **B1** — Strong invariance + conditional tubular erosion (corrected `02`).
- **B2** — Restricted proximal-normal composition (corrected `03`).
- **B3** — Finite-architecture exact-tube Operator II recursion (corrected `04`).
- **B4** — Sampled/RFDE/hybrid kernel chain, information-state kernel, sample-and-hold comparison (corrected `08`; A002 `thm:sampled-viability–thm:sampled-limit`).
- **B5** — Projectability criterion, fibre obstruction, spatial aggregation identity, support-saturated reduction, local-horizon bracket, Halanay small-gain (corrected `09`; A002 `thm:projectability`, `thm:coarse-graining`, `thm:horizon`, `thm:small-gain`).
- **B6** — Conservation, moiety positivity, noncompensation, Farkas substitution, observation fibres (corrected `07`; A002 `thm:conservation`, `thm:farkas`, `thm:observation`).
- **B7** — Operator I judgment hierarchy, Propositions 1–8, non-implications (corrected `01`).

- **B8** — A001 selected Operator I audit: recovery idempotence, common-action obstruction, delayed-information obstruction, adversarial exit (corrected 06).

```
                    B7 hierarchy ─────────────┐
                    B8 obstructions ──────────┤
                    B4 sampled kernels ───────┼──>  T2 closed-loop bridge (R02)
──┬──> institutional
                              ^               |
|    operationalization
                             |              └──>  T8 typed hierarchy maps (R08)
┘    (Wave 3, empirical)
        B1 erosion ───────────┼──>  T3 viability-diagnostic bridge (R03) ──>
empirical falsification
              ^               |               |
design (Paper 5)
             |               |              └──>  T4 domain admission (R04) ──>
domain classification
        B2 composition ───────┼──>  T5 restricted assume-guarantee (R05) ──┬──>
T6 aggregation (R06)
             ^               |                                       |      └─>
spatial/stage (A023)
              |               |                                          └─>
nonlinear small-gain [OPEN]
        B5 projectability ────┤
              |               └──>  T6 aggregation & memory (R06)
              |
        B3 Operator II ──────────>  T1 transformation false positives (R01) ──>
Paper 1 gate
              |
              └──────────────────>  T7 intergenerational recursion (R07) ──>
obligations/architecture change

        B6 conservation+noncompensation ──>  T9 boundary theorem (R09)  <──
consumes: B5,B7, R01,R03,R05,R06
                                             (synthesis; needs all U-part and
M-part witnesses)
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Adjacency (edge = “unblocks” / “is prerequisite for”):

From	To	Edge content
B7, B8, B4	**T2**	judgment quantifiers; common-action obstruction; exact/conservative filter pattern
T2	T8	lift/projection typed maps instantiated on the bridge’s information states
B1, B8, B5	**T3**	erosion lemma; adversarial-exit; local-horizon bracket (band-necessity)
B5, T2	**T4**	projectability for reduction claims; policy/information map typing
B1, B2	**T5**	erosion machinery; joint-feasibility pattern

From	To	Edge content
B5, B1, T5	**T6**	fibre criterion; erosion conversion for approximate closure; scale composition
B3, B6	**T7**	reset/translation semantics; obligation typing
B3	**T1**	exact-tube predecessor is the object whose endpoint-only weakening is refuted
B6, B7, B5 + R01, R03, R05, R06	**T9**	universal-part witnesses; model-class counterexample pairs

Critical path (longest prerequisite chain): **B1 → T3 → T9** and **B4 → T2 → T8**. The synthesis target T9 is a sink: it can only be closed after the M-part witnesses exist, which is why this review proves them first (R01, R03, R05, R06) and closes T9 last (R09).

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3. Verdict table for the docket targets (docket task 3)

Legend: verdicts as demanded by the master prompt (`already proved` / `repairable` / `false` / `classical but useful` / `genuinely new`), possibly compound. "Record" points to the full 17-field result file.

Target	Verdict	Basis	Record
T1 typed transformation beyond reachability	**classical but useful + genuinely new (negative results)**: the exact-tube backward recursion is standard robust dynamic programming (packet's own novelty gate says the same, `control/03` §2A.4); the endpoint-only and aggregate transformation tests are now provably unsound (new false-positive theorems with a sharp characterization: endpoint tests are sound only in the degenerate automatically-invariant case)	B3 + R01 new proofs	R01

Target	Verdict	Basis	Record
T2 observation–assessment–implementation bridge	**repairable → proved** (this review, at the sampled exact/conservative-filter level): one closed-loop theorem with exact quantifiers covering measured/hidden disturbance, exact/conservative filter, prescribed/realized action, latency/held commands, all-branches implementation; common-action counterexample retained; conservative filter proved sound but incomplete	B4, B7, B8	R02
T3 viability–diagnostic bridge	**repairable → proved (assembled)**: certificate trichotomy (outer soundness via adversarial exit; inner certificate via erosion conversion; otherwise descriptive); compactness closure of finite→infinite horizon; counterexamples: rate-band necessity (stock-to-rate margin fails by factor >4), aggregate margin is not a kernel	B1, B8, B5	R03

Target	Verdict	Basis	Record
T4 domain admission and projectability	**proved at the structural level**: exact-admission certificate theorem (necessary and sufficient map quintuple) + classification of groundwater/phosphorus/fisheries from the included sources with missing-field flags exactly matching the live error register; empirical calibration explicitly out of scope	B5, T2	R04
T5 composition beyond the first restricted theorem	**partially closed (new restricted theorems) + genuinely open remainder**: tubular assume–guarantee theorem with contract amplitude (Version A) and state-dependent contract tightening with a true linear gain operator and feasibility fixed point (Version B); counterexample: gain loop collapses the joint kernel to $\{0\}$; general nonlinear small-gain with nonconvex implementation stated precisely with all missing hypotheses, left open	B1, B2	R05

Target	Verdict	Basis	Record
T6 cross-scale aggregation and dynamic closure	**already proved at scoped level + genuinely new negative result**: exact characterization = projectability criterion (classical in substance, packet-proved); new: finite moment-closure impossibility for nonlinear (quadratic) field dynamics — no finite family of moments closes exactly, so exact coarse variables are necessarily memory-bearing; approximate closure carries O(κ)/variance defect with erosion conversion	B5, B1	R06
T7 intergenerational continuation	**repairable → proved** (set level): generation-indexed recursion separating fixed-specification viability from specification change; alternating-disjoint impossibility (continuous evolution cannot cross disjoint specifications — typed resets are necessary and sufficient rescue); nested-compact existence; monotone-obligation finite-horizon corollary; A001 Thm 14.2 impossibility retained	B3, B6	R07

Target	Verdict	Basis	Record
T8 robust–epistemic–chance hierarchy	**classical but useful (propositions) + new completion**: typed lift/projection maps formalized; five converse counterexamples (controlled⇏robust; chance-p⇏robust; support-mismatch; epistemic emptiness with nonempty physical kernel; implementation-branch monotonicity reversal)	B7, B8	R08
T9 general-theory boundary theorem	**proved (new as a stated theorem)**: Part U — the exact list of universal consequences of the structural axioms (conditional conservation, monotonicity calculus, noncompensation-in-domain, status/interface axioms); Part M — six independence results: for each of delay effect, substitution, dynamic aggregation closure, diagnostic causality, local-bifurcation/global-safety, and information monotonicity, two axiom-consistent instantiations with opposite truth values	B5, B6, B7 + R01, R03, R05, R06	R09

No docket target received the verdict `false` as a whole; the `false` verdicts apply to specific claims inside targets (endpoint-only tests: false as soundness claims — R01; universal delay law, universal substitution, universal dynamic aggregation, universal diagnostic causality: false as axiom-level theorems — R09 Part M; general metric erosion for arbitrary closed sets: already rejected in corrected `02`).

(Continued in §4–§7 below.)

4. Universal axioms versus model-class mechanisms (docket task 6)

The separation below is the load-bearing discipline for the phrase “general theory”: the left column is what the typed axioms force in every admitted instantiation; the right column is what remains a mechanism of a model class, with the paired witnesses that prove non-universality (full statements and proofs in R09).

4.1 Universal consequences of the structural axioms (Part U of R09)

#	Universal statement	Status / proof anchor
U1	**Conservation is conditional on declared closure and moiety:** for any declared boundary and moiety vector in the left kernel of every active internal/jump matrix, the telescoping identity `L^Tx(t) − L^Tx(0) = ∫ L^TBφ ds + Σ L^TB^J β_j` holds for every locally finite execution	proved, A002 `thm:conservation` (accepted with the φ-notation correction, corrected `07`); re-proved from the axioms in R09 §U.1
U2	**Viability is set-, quantifier-, and class-conditional:** the monotonicity calculus (Robust ⊆ Controlled; fixed-policy ⊆ Robust; action expansion enlarges; disturbance expansion shrinks; safe-set expansion enlarges) holds in every aligned instantiation	proved, corrected `01` Props 1–5; R09 §U.2
U3	**Noncompensation is conditional on declared binding components:** outside a domain with a proved scalar certificate, componentwise failure is not erased; on the restricted linear domain, feasibility is exactly Farkas-separated (pathway-specific certificate, never an exchange rate)	proved, A002 domain-qualified noncompensation + `thm:farkas` (accepted, corrected `07`); R09 §U.3
U4	**Status monotonicity and interface explicitness:** no integration step strengthens a proof/evidence status; no cross-module transfer without an admitted mapping + contract	axioms 5–6 of `TCS-1.0` §9 (definitional); R09 §U.4

#	Universal statement	Status / proof anchor
U5	**Exact kernel recursions equal their judgments under stated hypotheses:** sampled robust kernel, held-tube kernel, information-state kernel characterize their judgments exactly when their compactness/continuity/exactness hypotheses hold	proved, corrected `08`; R09 §U.5

4.2 Model-class-dependent mechanisms (Part M of R09) — none is a theorem of the axioms

#	Claimed universal law	Refuting pair (both axiom-consistent)	Record
M1	"Delay destabilizes" / "delay is harmless"	`ẋ = −2x(t) − x(t−τ)`: delay-independent exponential stability for every τ (Halanay certificate, A002 `thm:small-gain`, proved) **vs.** `ẋ = −x(t−τ)`: unstable for every τ > π/2 (explicit characteristic-root witness `x(t) = e^{λt}`, Re λ > 0, constructed in R09)	R09 §M.1
M2	"Substitution is always possible" / "never possible"	feasible linear pathway (Farkas multipliers certify an exact allocation) **vs.** infeasible pathway (strict separation certificate `γ^Ts^req > α^Tx + β^Te`) — both inside the same axiom system with different endowments	R09 §M.2 (A002 `thm:farkas` witnesses)

#	Claimed universal law	Refuting pair (both axiom-consistent)	Record
M3	“Coarse variables close dynamically”	identical-patch spatial system: mean dynamics exactly closed on the diagonal (variance ≡ 0 invariant) **vs.** two-patch heterogeneous system: fibre obstruction — same mean, different mean-derivatives, no autonomous mean closure; strengthened by R06: no finite moment family closes for quadratic field dynamics	R09 §M.3; R06 Thm 3
M4	“Diagnostics are causal certificates”	observation-fibre criterion: safety-crossing fibres obstruct any observation-only certificate (A002 `thm:observation`, proved) **vs.** injective observation on the declared domain: exact certification; plus R03: stock-to-rate margin fails by unbounded factor without a rate-persistence band	R09 §M.4; R03 Thm 2
M5	“Local bifurcation ⇒ global safe-set change”	fold in an uncoupled coordinate leaves the viability kernel of an independent safe set unchanged (explicit 2-D witness, R09) **vs.** A018 C3/C4-class systems where the fold interacts with the constraint set (source-stated status, not promoted here)	R09 §M.5

#	Claimed universal law	Refuting pair (both axiom-consistent)	Record
M6	"More information/implementation is always better"	refinement monotonicity holds only through typed lift maps (corrected `01` Prop 6) **vs.** all-branches implementation enlargement shrinks safety (Prop 7 reversal; one-step witness in R08 Ex (e))	R09 §M.6; R08

4.3 Reading of the separation

The theory's universality is exactly of the form **"conditional laws + obstruction calculus"**: conservation, viability, noncompensation, kernel recursion, and status/interface discipline are universal as conditional statements with typed hypotheses; every mechanism that would make the theory predictive without a model — delay effects, substitution elasticities, aggregation closure, diagnostic causality, bifurcation-to-collapse links — fails at least one independence witness. This is the honest content of the phrase "general theory" under TCS typing and is stated as a theorem (not a narrative) in R09.

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5. Source-level unity versus publication-level dependence (docket task 7)

The master prompt requires preventing the (real) source-level unity of the corpus from becoming (unsupported) publication-level dependence. The audit against `control/05_publication_architecture.md` (federated orchestration, citation-closure rules) and the corrected records yields:

1. **The seam that had to be exact is exact.** The A018 ledger→dynamics seam is closed by an `EXACT_SPECIALIZATION` (single-resource deficit identity $D = qEN - R = -\dot{N}$) plus an explicit `REJECTED_MAPPING` for the dynamic reduction (corrected `05`). Paper 3/Paper 4 dependence is therefore contextual, not load-bearing — the refereeability test in that record is satisfied. This review adds nothing to that seam and flags one enforcement obligation: Paper 4 must carry the open-projection accounting paragraph verbatim (corrected `05`, "Open-projection accounting").

2. **Two new load-bearing interfaces are now proved and must be cited as theorems, not narratives.** The closed-loop bridge (R02) is the first interface theorem connecting observation, filter, prescription, and implementation in one quantifier chain. Papers 1 and 5 currently describe this chain architecturally (`TCS-1.0` §3); after R02 they may cite a theorem, but only with its exact hypotheses (compact held-action sets, declared implementation correspondence, conservative-update soundness). The diagnostic bridge (R03) is the analogous interface for Paper 5's falsification design: margin diagnostics may be cited as outer/inner certificates only with the adversarial-exit or erosion hypotheses attached.

3. **One tempting dependence is now formally blocked.** Because endpoint-only transformation tests are unsound (R01), no paper may present an architecture transformation as safe on the basis of successor-set membership alone; and because aggregate tests are unsound under noncompensatory constraints (R01 Thm 2), no paper may certify transition safety from an aggregate/projection tube. Both prohibitions are now

theorems, not editorial cautions — they belong in Paper 1's transformation section and in the monograph's composition chapter.

4. **The composition gate remains two-theorem-wide.** Until the nonlinear small-gain/open problem of R05 §Field 16 closes, cross-module safety claims may cite only the restricted proximal-normal theorem (corrected `03`) and the tubular assume–guarantee theorem (R05) — the latter only with its erosion/gain constants recorded per the `erosion_triple` field proposed in §1.1 GAP-3. Any other composition claim stays `UNRESOLVED`.

5. **Empirical and novelty dependencies are unchanged and remain external.** This packet is self-contained for the mathematics (per its own report) but not for bibliographic novelty adjudication or calibration data. Every novelty field in R01–R09 says precisely what internal evidence supports and what external check is outstanding; the research plan (§7) keeps both outside the critical path of the theorem work.

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6. Minimum set of new theorems justifying "general theory" (docket task 8)

The claim "general theory of sustainability" under TCS typing is justified — without implying universal delay, substitution, aggregation, or empirical laws — by exactly the following closure set (bases B1–B8 are already proved in the packet; the numbered items are the new closures, each proved in this review):

1. **Closed-loop governance bridge (R02).** One theorem carrying the full causal chain physical state → observation → information state → prescription → realized action → trajectory with all docket distinctions enforced. Why minimum: without it the theory describes modules but cannot state a governance guarantee.
2. **Scope boundary theorem (R09).** The exact Part U / Part M split. Why minimum: it is what licenses the word "general" honestly — universal conditional laws plus proved non-universality of the predictive claims.
3. **Viability–diagnostic bridge (R03).** Soundness/completeness/descriptive trichotomy + erosion conversion + horizon closure. Why minimum: it connects the normative kernel language to anything empirically testable, which the docket's T3 and Paper 5 require.
4. **Domain admission certificate (R04).** Necessary-and-sufficient map quintuple + classification of the three domain modules. Why minimum: "general" across domains needs an admission test, not verbal analogy; the certificate is the test.
5. **Restricted assume–guarantee composition (R05) + aggregation non-closure (R06).** The positive restricted composition rule with true gain operator, and the negative moment-closure result. Why minimum: together they state exactly when modules compose and when scales do not close — the two honesty boundaries of any cross-scale theory.

Everything else in the docket is either already proved (bases), a refinement that does not change the claim structure (R01's false-positive theorems sharpen Operator II's necessity; R07 separates specification change; R08 completes converse counterexamples), or remains explicitly conditional (nonlinear small-gain, variable-event hybrid kernels, stochastic filter exactness, empirical calibration). **With items 1–5 proved, the phrase "general theory" is defensible; without any one of them it is not:** dropping R09 leaves "general" unbounded; dropping R02 leaves governance unhinged; dropping R03/R04 leaves the theory untestable and domain-bound; dropping R05/R06 leaves cross-scale claims unaudited.

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7. Dependency-ordered research plan (closing requirement of the master prompt)

Ordering rule per `control/03_dependency_plan.md` (prerequisite depth, downstream unblocking, validity risk, feasibility, value). Waves A–C below are complete in this review; waves D–F are the remaining programme.

Wave A — negative/structural closures (no unmet prerequisites; done here)

- A1. Endpoint-only and aggregate transformation tests: unsoundness theorems + degenerate-case characterization (**R01**). Unblocks: Paper 1 transformation-section discipline.
- A2. Hierarchy completion: typed lift/projection maps + five converse counterexamples (**R08**). Unblocks: Paper 2 hierarchy section finalization.
- A3. Boundary theorem Part M witnesses (**R09**, together with waves B–C inputs). Unblocks: monograph scope chapter.

Wave B — positive bridge closures (prerequisites: B1, B4, B8; done here)

- B1*. Closed-loop observation→assessment→prescription→implementation theorem + conservative-filter soundness/incompleteness + common-action necessity (**R02**). Unblocks: institutional operationalization (wave E), Paper 5 closed-loop design.
- B2*. Diagnostic bridge: trichotomy + erosion conversion + compactness horizon closure + rate-band and aggregate-margin counterexamples (**R03**). Unblocks: falsification design, T4's diagnostic fields.

Wave C — composition, scale, generation (prerequisites: B1, B2, B3, B5; done here)

- C1*. Tubular assume–guarantee theorem, Versions A and B, with linear gain feasibility + collapse counterexample (**R05**). Unblocks: T6, A023 spatial branch, polycentric composition docket.
- C2*. Moment-closure impossibility + memory-necessity + approximate-closure erosion conversion (**R06**). Unblocks: A023, cross-scale claims in the monograph.
- C3*. Generation recursion + specification-change separation + alternating-disjoint impossibility + nested-compact existence (**R07**). Unblocks: intergenerational applications, `TCS-1.1` specification-path type.
- C4*. Domain admission certificate + classification of groundwater/phosphorus/fisheries (**R04**). Unblocks: wave E case selection.

Wave D — conditional extensions (prerequisites: waves B–C; open, precisely stated)

- D1. Nonlinear small-gain / assume–guarantee with nonconvex implementation and shared controls beyond R05's tubular class (R05 field 16: all missing hypotheses listed). Parallelizable once R05 constants are instantiated on one application pair.
- D2. Measurable/continuous selector regularity for R02's witness correspondence (currently: arbitrary selector, set-valued existence — the packet's standing selector caution, corrected `08` §1).
- D3. Stochastic filter exactness + chance-viability support alignment instances (QF-2 guard; needs a declared law).
- D4. Variable-event delayed-hybrid kernel (A002 conjecture; unchanged by this review).

Wave E — empirical instantiation (prerequisites: R02, R03, R04; external data)

- E1. Groundwater vs. phosphorus readiness comparison per `control/03` wave 3 (groundwater preferred unless the matrix fails); admission certificate instantiated per R04 with the module's open fields (error register B.5 items) closed first.

- E2. Institutional/distributive operationalization on the selected case.

Wave F — audit and publication (prerequisites: stable theorem set; external literature)

- F1. Global novelty map: R01–R09 novelty fields are pre-populated with internal evidence and outstanding external checks; the audit must verify each against robust DP/reachability, viability, hybrid-systems, small-gain, and moment-closure literatures.
- F2. Paper-1 independent-result gate: R01 + R05 give the false-positive/negative and restricted-positive content that the gate sought beyond the standard recursion; decision on sufficiency is editorial, after F1.
- F3. Per-paper artifact manifests, `TCS-1.1` migration (§1.4 diff), drafting per `control/05`.

Parallelism: A1–A3 and B1–B2 are independent of C1–C4; within C, C1–C4 are independent. D1–D4 run in parallel after C. E requires B* and C4* only. F1 can start as soon as any wave-B/C record stabilizes. **Explicitly conditional forever (never silently promoted):** D1 in full generality, D4, empirical calibration claims, and any bibliographic novelty claim before F1.

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8. What was checked, what was proved, what remains open — one-paragraph summary

The packet's controlling objects were audited (§1): TCS-1.0 is sound with five additive gaps and four quantifier guards proposed for TCS-1.1. The dependency graph for T1–T9 was built (§2) and every target received a verdict with a full 17-field record (§3, files R01–R09): the exact-tube Operator II recursion is classical, and its endpoint-only and aggregate weakenings are now provably unsound (R01); the observation–assessment–prescription–implementation bridge is proved as one closed-loop theorem with exact quantifiers and a common-action necessity counterexample, including the sound-but-incomplete conservative-filter calculus (R02); the viability–diagnostic bridge is proved as a certificate trichotomy with erosion conversion, horizon closure, and two necessity counterexamples (R03); domain admission is a necessary-and-sufficient certificate theorem that classifies the three included domain modules without verbal analogy (R04); composition gains a genuine restricted assume–guarantee theorem with a linear gain operator and feasibility fixed point, with the general nonlinear small-gain problem stated precisely and left open (R05); aggregation is exactly characterized by projectability, and finite moment closure is proved impossible for quadratic field dynamics, making memory-bearing closure a theorem rather than a slogan (R06); intergenerational continuation is a generation-indexed recursion separating fixed specification from specification change, with the disjoint-alternation impossibility proving typed resets necessary (R07); the judgment hierarchy is completed by typed comparison maps and five converse counterexamples (R08); and the general-theory boundary theorem states exactly which claims are universal consequences of the typed axioms and proves six independence results for the predictive claims that are not (R09). The minimum closure set justifying "general theory" is identified (§6) and the remaining programme is dependency-ordered (§7), with novelty fields honestly limited to internal evidence pending external literature access.

Result Record R01 — Docket T1: False Positives of Endpoint-Only and Aggregate Transformation Tests

Field 1 — Result ID and target docket item

R01 (three results: R01.Thm1 endpoint-only unsoundness; R01.Thm2 aggregate unsoundness; R01.Prop3 divergence mechanism). Target: **T1** ("a theorem proving false positives of endpoint-only/aggregate transformation tests"). Strengthens the proved exact-tube Operator II predecessor (corrected 04).

Field 2 — Verdict

- R01.Thm1, R01.Thm2: **proved** (new negative results).
- R01.Prop3: **proved** (mechanism generalization).
- The underlying exact-tube recursion itself: **classical but useful** (robust backward reachability; consistent with the packet's own provisional novelty answer, control/03 §2A item 4).

Field 3 — Exact statement

All statements use the Operator II data of corrected 04 §1 (finite review times $0=t_0<\dots<t_m=T$, meta-actions A_k, disturbance sets D_k, exact tubes $Tube_k$, exact successor sets $Succ_k$, transition-safe sets S_k, terminal set G), with the robust predecessor

$$RPre_k(W) = \{(q,x) \in S_k : \exists a \in A_k\ \forall d \in D_k : Tube_k(q,x,a,d) \subseteq S_k \text{ and } Succ_k(q,x,a,d) \subseteq W\}$$

and its **endpoint-only weakening** $RPre^e_k(W)$ defined by deleting the tube clause.

R01.Thm1 (endpoint-only false positives). There exist admissible Operator II data with Lipschitz right-hand side, compact action and disturbance sets, unique forward-complete solutions, and a nonempty exact-tube kernel W_0, such that:

1. $W_0 = \varnothing$ (no state is robustly transformable), while
2. the endpoint-only recursion returns $W^e_0 = S_0 \neq \varnothing$.

Consequently the endpoint-only test certifies robust transformability for states at which every causal policy violates transition safety within the first interval. Moreover, the endpoint-only recursion is sound for given data only if no accepted state admits a mid-interval excursion out of S_k — i.e. only if the tube clause never binds; it is degenerate exactly when within-interval transition safety is automatic.

R01.Thm2 (aggregate false positives). Let $P: X \to Y$ be an aggregate/projection map and let the aggregate transformation test replace $Tube_k \subseteq S_k$ by $P(Tube_k) \subseteq P(S_k)$ (and endpoints correspondingly). There exist admissible data with a noncompensatory safe set (product of component constraints) for which:

1. every trajectory satisfies $P(x(t)) \in P(S)$ for all t (the aggregate test passes on every branch, forever), while
2. the componentwise viability kernel is empty — no admissible policy keeps the state in S for any positive time.

R01.Prop3 (divergence mechanism). In the witness of Thm1 the failure is not an artifact: if, in some direction e, the safe set S_k has width $w > 0$ and the unmeasured-disturbance branch divergence in direction e satisfies $\langle e, x^{d_1}(t) - x^{d_2}(t)\rangle \geq \beta t$ for all policies and some $\beta > 0$, then no causal policy can keep the tube in S_k beyond time w/β, whatever the endpoint behavior. Endpoint-only tests are blind to exactly this quantity.

Field 4 — State and phase space

Thm1: one architecture q, $X = \mathbb{R}$ (finite-dimensional ODE class; phase state is the physical coordinate alone). Thm2: $X = \mathbb{R}^2$, safe set $S = [0,1]\times[0,1]$, aggregate $P(x_1,x_2) = x_1 + x_2$, aggregate space $Y = \mathbb{R}$. Both are sampled systems with within-interval continuous evolution and review-time meta-actions, per TCS-1.0 §2.2 (sampled systems use flows between review times plus held commands).

Field 5 — Quantifier order and information pattern

Exactly the Operator II order of corrected 04 §1: $\exists a \in A_k\ \forall d \in D_k$ (action chosen before disturbance; disturbance unmeasured within the interval; full phase state observed at review times). Thm1's hard direction uses the adversarial reading of $\forall d$ (both disturbance branches must be safe simultaneously under one action). No disturbance-observing control is admitted (master prompt, non-negotiable rule 1).

Field 6 — Assumptions, including existence/completeness

- f Lipschitz in (x,u) uniformly in d; compact U, D; linear growth ⇒ unique, forward-complete solutions for held actions and measurable disturbances (satisfied by both witnesses).
- $Tube_k$, $Succ_k$ are the exact all-branch sets of the declared solution concept (as required by corrected 04 §1 items 3–4; conservative outer tubes would only enlarge the false positives).
- Thm2's safe set is a product of component intervals (noncompensatory registry, TCS-1.0 §2.5); no Ω-authorized substitution (axiom 4).

Field 7 — Mapping type

COUNTEREXAMPLE_OR_LIMIT for the soundness of endpoint-only and aggregate tests (they are REJECTED_MAPPING instruments); the results support the TRANSFORMATION mapping type of the exact-tube predecessor by proving its tube clause cannot be dropped or projected.

Field 8 — Self-contained proof

Proof of R01.Thm1

Data. Two stages: $t_0 = 0 < t_1 = 1 < t_2 = 2$; one architecture. Dynamics $\dot{x} = u + d$ with $u \in U = [-1,1]$ (meta-action = any causal within-interval rule measurable in t, constant or not) and unmeasured disturbance $d(t) \in D = \{-1, +1\}$ (constant on the interval — the declared disturbance class D consists of the two constant branches; measurability and admissibility are immediate). Solutions are unique for each $(u(\cdot), d)$ and forward complete.

Safe sets: $S_0 = [-1/4, 1/4]$ (binding), $S_1 = \mathbb{R}$ (non-binding final interval). Terminal destination $G = [-1, 1]$.

Step 1 — exact stage-1 predecessor. $W_1 = RPre_1(G)$ requires $\exists u(\cdot)\ \forall d \in \{\pm1\}$: $Succ_1(x, u, d) \subseteq [-1,1]$ (the tube clause against $S_1 = \mathbb{R}$ is void). Choose the constant rule $u \equiv -x$; then $\int_0^1 u = -x$ and the endpoints are $x - x + d \in \{-1, +1\} \subseteq G$, for **every** $x \in \mathbb{R}$. Hence $W_1 = \mathbb{R}$.

Step 2 — exact stage-0 predecessor is empty. $W_0 = \{x \in S_0 : \exists u(\cdot)\ \forall d \in \{\pm1\}: Tube_0(x,u,d) \subseteq S_0 \text{ and } Succ \subseteq W_1 = \mathbb{R}\}$. The successor clause is void ($W_1 = \mathbb{R}$), so W_0 is exactly the set of states with a robustly tube-safe action. Let $x^{+}(t)$, $x^{-}(t)$ be the two branch trajectories under one rule $u(\cdot)$ (the rule cannot depend on the unmeasured d). Then $x^{+}(t) - x^{-}(t) = \int_0^t [(u+1) - (u-1)]\, ds = 2t$. Both branch trajectories must lie in S_0, an interval of width 1/2. Two points 2t apart both lie in an interval of width 1/2 only if $2t \le 1/2$, i.e. $t \le 1/4$. For $t > 1/4$ this is impossible, **for every** $x \in S_0$ and every rule $u(\cdot)$. Hence no robustly tube-safe action exists anywhere in S_0, and $W_0 = \varnothing$: no state is robustly transformable.

Step 3 — endpoint-only predecessor is full. $W^e_0 = \{x \in S_0 : \exists u(\cdot)\ \forall d: Succ_0(x,u,d) \subseteq W_1 = R\}$. The successor clause is void, so $W^e_0 = S_0 \neq \varnothing$ (any rule witnesses, e.g. $u \equiv 0$, whose endpoints $x \pm 1$ are trivially in R).

Step 4 — degeneracy claim. The same computation shows the general point: whenever the successor sets $Succ_k$ can be steered into the next-stage set while the branch tubes diverge faster than the safe-set width absorbs, the endpoint-only recursion accepts and the true recursion rejects. Soundness of the endpoint-only recursion for given data is equivalent to: for every accepted (q,x) and every accepted action, $Tube_k \subseteq S_k$ — i.e. the tube clause never binds on the accepted set. □

Remark (why the packet's exact-tube clause is not dispensable). Step 2 is the operational content: the endpoint test measures dispersion at review times, which the controller can cancel ($u \equiv -x$ re-synchronizes endpoints exactly); the tube condition measures uncontrolled within-interval dispersion ($2t$), which no causal rule can cancel. These are different quantities; the theorem proves conflating them is unsound, not merely imprudent.

Proof of R01.Thm2

Data. $X = R^2$, $S = [0,1]\times[0,1]$, dynamics $\dot{x}_1 = c + u,\ \dot{x}_2 = -(c + u)$, with $u \in [-1,1]$ and constant $c > 1$ (e.g. $c = 2$). Aggregate $P(x_1,x_2) = x_1 + x_2$, $P(S) = [0,2]$.

Step 1 — the aggregate is exactly constant. For every admissible rule and every branch, $d/dt\ (x_1 + x_2) = 0$, so $P(x(t)) \equiv P(x(0)) \in [0,2]$ for every trajectory starting in S. Hence $P(Tube) \subseteq P(S)$ and $P(Succ) \subseteq P(S)$ hold on every branch for all time: the aggregate transformation test passes unconditionally.

Step 2 — the componentwise kernel is empty. Since $c > 1$, the velocity set is $c + u \in [c-1,\ c+1] \subset (0,\ \infty)$: x_1 is strictly increasing at rate at least $c-1$ and x_2 strictly decreasing at rate at least $c-1$, for **every** admissible control. From any $x \in S$, either x_1 reaches 1 or x_2 reaches 0 within time $1/(c-1)$. Both faces belong to ∂S and the velocity points strictly outward on them ($\dot{x}_1 > 0$ on $\{x_1 = 1\}$, $\dot{x}_2 < 0$ on $\{x_2 = 0\}$), so every trajectory exits S in finite time and no state is viable: $Viab(S) = \varnothing$ (robustly and even existentially, since the sign of $c+u$ is control-independent).

Step 3 — conclusion. The aggregate test certifies safety forever on every branch; the true noncompensatory kernel is empty. By the noncompensation axiom (TCS-1.0 §9, axiom 4), the componentwise failure is not erased by the aggregate surplus: the two flows compensate in the aggregate while both components are destroyed — precisely the structure axiom 4 forbids aggregating away. □

Proof of R01.Prop3

Let e be a unit direction and suppose $S_k \cap (x + Re)$ has width w (i.e. $\sup\{\langle e,\ y-x\rangle : y \in S_k,\ \langle e',\ y-x\rangle = 0\ \forall e'\perp e\} - \inf\{\dots\} = w$ — concretely, the extent of S_k along e). If two admissible disturbance branches satisfy $\langle e,\ x^{d_1}(t) - x^{d_2}(t)\rangle \geq \beta t$ for $t \in [0,\ t_{k+1}-t_k]$ under **every** causal rule (divergence is policy-independent — as in Step 2 of Thm1, where the divergence $2t$ is the integral of the disturbance difference), then at any time t with $\beta t > w$ the two branch states cannot both lie in S_k. Hence every tube exits S_k by time w/β. The endpoint-only test never evaluates the pair $(w,\ \beta)$; the exact-tube test does. □

Field 9 — Counterexample showing necessity or failure outside scope

The two witnesses are the counterexamples (Fields 3, 8). Additionally: if D_k is a singleton (no unmeasured disturbance) **and** within-interval rules are restricted to held-constant controls with convex safe sets, endpoint tests can be sound (the tube is then the chord between state and endpoint, contained in a convex `S_k` whenever endpoints are); this identifies exactly the two hypotheses whose absence breaks soundness: branch multiplicity and non-convex or time-varying tube geometry. Both hypotheses are generic in the programme's sampled-governance setting (corrected `08`), so the failure is structural, not pathological.

Field 10 — Interface producer/consumer contract

- **Producer:** this record (negative soundness results for weakened Operator II tests).
- **Consumer:** Paper 1 (transformation section: may cite that endpoint/aggregate certification is provably insufficient, so the exact-tube clause is mandatory); the monograph composition chapter; any downstream module tempted to certify an architecture change from endpoint snapshots (e.g., A018-style ledger snapshots between regimes).
- **Type/unit map:** none needed (pure metatheorem about the Operator II data).
- **Failure condition:** the results are revoked only if the Operator II data of corrected `04` change (e.g., a solution concept in which tubes are determined by endpoints — true only for convex `S_k`, singleton `D_k`, held convexifying rules, per Field 9).

Field 11 — Error, horizon, and safety erosion for approximations

Not an approximation result: the theorems are exact. For conservative outer tubes the packet's Corollary 3 (corrected `04`) remains controlling: outer tubes give inner certificates; R01 shows that no tube information at all gives no certificate — the two results bracket the information requirement: **endpoints alone: unsound (R01.Thm1); exact tubes: exact (corrected `04`); outer tubes: sufficient (corrected `04` Cor. 3)**.

Field 12 — Selector and implementation regularity

None claimed or needed: the negative results hold for arbitrary causal rules, so they hold a fortiori for any measurable/Lipschitz/regular selector subclass. (This is the pleasant quantifier direction: refuting all rules refutes regular subfamilies.)

Field 13 — Stochastic/hybrid/RFDE qualifications

- The witnesses are deterministic-disturbance sampled ODE systems (the weakest class), so the negative results transfer to every richer class that contains it (RFDE with zero delay, hybrid without events, stochastic degenerate laws) — containment mappings are identity embeddings, `EXACT_SPECIALIZATION`.
- For genuinely stochastic branches, β in Prop3 becomes the branch-divergence rate of the support; chance-constrained endpoint tests fail identically whenever the support has two branches — no new phenomenon.

Field 14 — Novelty status with exact references

- Internal evidence: the packet's provisional novelty answer already assesses the exact-tube recursion as "mathematically standard" (`control/03` §2A item 4) and its acceptance criterion for T1 explicitly demands "a theorem proving false positives of endpoint-only/aggregate transformation tests" (`11_OPEN_THEOREM_DOCKET.md` T1) — i.e., the packet itself flags these negative results as the missing content; this record supplies them.
- The divergence argument (Prop3) is elementary; the packaging (soundness degeneracy of endpoint-only robust predecessor recursions) is, to the packet's internal knowledge, new, but **external literature verification against robust reachability / hybrid-safety verification literature is outstanding** (packet self-containment report caveat). No external bibliographic claim is made here.

Field 15 — Publication destination

Paper 1 (architecture/transformation section — as a boxed impossibility proposition with the one-dimensional witness); Paper 2 (theorem atlas, counterexample register). Monograph: composition/transformation chapter.

Field 16 — Remaining obligations and revocation triggers

- Obligations: external novelty check (Field 14); a two-architecture instantiated transformation example for Paper 1 (per corrected `04` §9 item 4) should carry the endpoint-false-positive example as its motivation.
- Revocation triggers: change of the Operator II data conventions (Field 10); discovery of an external theorem identical in scope (would demote novelty, not truth).

Field 17 — Machine-readable dependency edges


```
{
  "result_id": "R01",
  "target": "T1",
  "depends_on": ["corrected_theorems/04_operator_II_transformation_candidate.md
  (exact-tube predecessor, data conventions)"],
  "unblocks": ["Paper 1 transformation discipline", "R09.M3 (aggregate witness
  reuse)", "R03.Thm3 (aggregate margin counterexample reuses R01.Thm2 system)"],
  "status": {"R01.Thm1": "proved", "R01.Thm2": "proved", "R01.Prop3": "proved"},
  "mapping_type": "COUNTEREXAMPLE_OR_LIMIT",
  "novelty": "internal-new; external check outstanding"
}
```

Result Record R02 — Docket T2: The Observation–Assessment–Prescription–Implementation Closed-Loop Bridge

Field 1 — Result ID and target docket item

R02 (R02.Thm1 closed-loop robust institutional viability; R02.Lem2 conservative-filter soundness; R02.Prop3 conservative incompleteness; R02.Prop4 common-action necessity; R02.Cor5 measured-disturbance variant; R02.Cor6 eroded closed-loop safety). Target: **T2** (“prove the observation → assessment → prescription → implementation bridge”).

Field 2 — Verdict

Repairable → proved (this record, at the sampled, exact/conservative-filter, finite-horizon level). The five distinctions demanded by the docket — measured vs. hidden disturbance, exact vs. conservative filter, prescribed vs. realized action, latency and held commands, all-branches vs. existential implementation — are each enforced by a typed hypothesis or quantifier of Thm1, and the acceptance criterion (“one closed-loop theorem with exact quantifiers and a common-action counterexample”) is met in full.

Field 3 — Exact statement

Data

1. Review times $0 = t_0 < t_1 < \dots < t_N = T$; interval $I_k = [t_k, t_{k+1}]$.
2. **Physical layer:** closed state space $X \subseteq R^n$, closed safe set $K \subseteq X$; plant $f: X \times U \times D \to R^n$ with U, D compact, f Lipschitz in (x,u) uniformly in d, linear growth. For every held $u \in U$ and measurable $d: I_k \to D$ the initial-value problem $\dot{x} = f(x,u,d(t))$ has a unique forward-complete solution (denoted $\varphi_{u,d}$).
3. **Observation layer:** deterministic set-valued observation map $O: X \to Y$ (closed values; the review-time observation is $Y_k = O(x(t_k))$).
4. **Assessment layer:** information states are pairs (C, c) with $C \subseteq X$ closed and $c \in U$ (the deployed/held command). The conservative update of a compatible set under a realized-action **set** $\tilde{U} \subseteq U$ is $\Phi(C, \tilde{U}) := \{x' \in X : \exists x \in C, \exists u \in \tilde{U}, \exists d \in D: x' = \varphi_{u,d}(x)(t_{k+1})\}$, and the post-observation conservative state is $\Phi(C, \tilde{U}) \cap O^{-1}(Y')$.
5. **Prescription layer:** compact command set U^{cmd}; policies map information states to commands.
6. **Implementation layer:** declared non-strategic correspondences (declared independently of the policy class, per schema-audit item CIRC-3): $I: U^{cmd} \times U \times K(X) \to$ nonempty compact subsets of U (realized actions), and $DeP: U^{cmd} \times U \to$ nonempty compact subsets of U (deployed command at the next review; $\{u^{cmd}\}$ = no latency, $\{c\}$ = one-step full latency, mixtures allowed).
7. **Certificate family:** $V \subseteq \{(C,c) : C \subseteq K \text{ closed nonempty}, c \in U\}$, **downward closed** in C ($(C,c) \in V$, $\varnothing \neq C' \subseteq C$ closed $\Rightarrow (C',c) \in V$).
8. **True compatible sets:** B_0 declared prior with $B_0 \subseteq C_0$; $B_{k+1} = \Phi(B_k, \{u^{real}_k\}) \cap O^{-1}(Y_{k+1})$ along the true branch.

Regulation condition (REG)

$(C,c) \in V$ satisfies (REG) if there exists $u^{cmd} \in U^{cmd}$ such that, writing $\tilde{U} := I(u^{cmd}, c, C)$:

- **(i) tube clause:** for every $u^{real} \in \tilde{U}$, every $x \in C$, every $d \in D$: $\varphi_{u^{real},d}(x)(t) \in K$ for all $t \in I_k$;
- **(ii) successor clause:** for every $c' \in DeP(u^{cmd}, c)$ and every observation value Y' with $\Phi(C, \tilde{U}) \cap O^{-1}(Y') \neq \varnothing$: $(\Phi(C, \tilde{U}) \cap O^{-1}(Y'), c') \in V$.

R02.Thm1 (closed-loop robust institutional viability)

If $(C_0, c_0) \in V$, $B_0 \subseteq C_0$, and (REG) holds at every $(C,c) \in V$, then there is a causal, observation-history-based prescription policy π such that for every true initial state $x_0 \in B_0$, every observation realization, every implementation branch $u^{real}_k \in I(\pi_k, c_k, C_k)$, every deployment branch $c_{k+1} \in DeP(\pi_k, c_k)$, and every disturbance realization $d_k \in D$ on each interval, the closed-loop trajectory exists, is unique on each interval, and satisfies

$$x(t) \in K \quad \text{for all } t \in [0, T].$$

Moreover the institution's own information state (C_k, c_k) remains in V and satisfies $B_k \subseteq C_k$ for all k, and π is computable from (C_k, c_k) alone.

R02.Lem2 (conservative soundness)

For any inclusion-monotone conservative update (in particular Φ above, in both arguments), $B_k \subseteq C_k$ for all k along every branch — conservative assessment never understates compatibility.

R02.Prop3 (conservative incompleteness)

There exist a plant, safe set, observation map, prior, and two admissible filters — one exact, one conservative (coarser observation processing) — with the same data, such that the exact filter satisfies the hypotheses of Thm1 while the conservative one violates (REG) through a common-action obstruction. Conservative assessment is therefore sound but incomplete: it can reject safe systems, never certify unsafe ones.

R02.Prop4 (common-action necessity)

If (C,c) is reachable and contains a state on ∂K, no informative observation arrives before the next prescription, and

$$\bigcap_{x \in C} \{u \in U : \text{every branch from } x \text{ under } u \text{ stays in } K \text{ on } I_k\} = \varnothing,$$

then no observation-based policy is robustly safe from (C,c): (C,c) lies in no certificate family satisfying (REG).

R02.Cor5 (measured-disturbance variant)

If the disturbance factors as $d = (d^m, d^h)$ with the measured component $d^m(t_k)$ revealed before the k-th prescription, then (REG) may be weakened: the witness command may depend on the revealed value, and clauses (i)–(ii) quantify over d^h only, conditionally on the revealed d^m. The conclusion of Thm1 holds conditionally on each realized measured sequence.

R02.Cor6 (eroded closed-loop safety)

If, additionally, the erosion-lemma hypotheses of corrected 02 Lemma 2 hold for K (two-sided tubular radius ρ, $C^{1,1}$ signed distance, normal correspondence) and the closed-loop velocity envelope has Lipschitz gain L_G and boundary margin α, then the tube clause (i) may be verified on the eroded set K_{-r} in place of K provided

$$L_G r + \Delta_\varepsilon \le \alpha, \quad 0 < r < \rho, \quad K_{-r} \neq \varnothing,$$

with $\Delta_\varepsilon = C\varepsilon + \mu$ the combined implementation/observation/model error budget ($C = L_u(L_\kappa a_o + a_u)$ per corrected 02; μ the plant-model error). The conclusion (true-state safety in K) then holds with that budget.

Field 4 — State and phase space

Extended sampled phase space per `TCS-1.0` §2.1: physical block $x \in X$; held-command block $c \in U$ (implementation/queue state); information block $C \subseteq X$ (compatible-set estimate — the `b` block in the canonical typing). The decision chain instantiated is exactly `TCS-1.0` §3: physical state → `O` → observation → conservative filter Φ → information state → π → prescription → `I/DeP` → realized action and next deployed command → plant. Model class: sampled ODE with held actions (`sampled_hybrid` without interior events, per the corrected `08` architecture).

Field 5 — Quantifier order and information pattern

The theorem's conclusion quantifier chain:

```
∃π (causal, observation-history-based)
 ∀x_0 ∈ B_0  ∀observations  ∀u^real branches  ∀deployment branches  ∀d ∈ D:
   trajectory exists uniquely and stays in K on [0,T].
```

The prescription precedes the realization of u^{real} and d on each interval (π_k is a function of (C_k, c_k), i.e. of history up to t_k); the realized action is chosen by the institutional correspondence (all-branches semantics: $\forall u^{real} \in I$), not by the policy — the master prompt's rule that one causal policy must be distinguished from disturbance-observing controls is enforced structurally: the policy cannot see d (except the measured component in Cor5, which is declared information, not control of the quantifier order) and cannot see u^{real} before prescribing.

Field 6 — Assumptions, including existence/completeness

Existence/uniqueness/forward completeness: hypothesis 2 (Lipschitz + linear growth + compact U,D). Nonempty `I`, `DeP` values (hypothesis 6) guarantee every implementation/deployment branch is well-defined. Nonemptiness of the conservative sets along the true branch is proved, not assumed (Step 3 of the proof). The certificate family's nonemptiness is the load-bearing hypothesis: (REG) at every $(C,c) \in V$ is the "assessment succeeds" assumption — Thm1 is deliberately conditional in the same sense as the packet's exact-filter theorems (corrected `08`: "assumes, rather than derives, exact compatible-state sets"), with the conservative variant made explicit so that the condition is verifiable with outer approximations.

Field 7 — Mapping type

`EXACT_SPECIALIZATION` of the `TCS-1.0` execution chain (§3) to the sampled class, plus `TRANSFORMATION`-free composition: no reduction between model classes is claimed. Prop3 is `COUNTEREXAMPLE_OR_LIMIT` for filter-exactness claims.

Field 8 — Self-contained proof

Proof of R02.Thm1

Define the closed loop inductively. Base: $(C_0, c_0) \in V$ (hypothesis). Step, given $(C_k, c_k) \in V$ and the true state $x(t_k) \in B_k$:

Choice of prescription. Let $\pi_k := u^{cmd}$ be any witness of (REG) at (C_k, c_k). This is a function of (C_k, c_k) only, hence causal in the observation/command history (see causality note below).

Realization. The implementation selects $u^{real}_k \in \tilde{U}_k := I(\pi_k, c_k, C_k)$; deployment selects $c_{k+1} \in DeP(\pi_k, c_k)$. Both are arbitrary within their declared correspondences (all-branches).

Safety on the interval. The true state $x(t_k) \in B_k \subseteq C_k$ (induction; base $B_0 \subseteq C_0$). Clause (REG)(i) at (C_k, c_k) applies to every $x \in C_k$, in particular to $x(t_k)$, with the realized $u^{real}_k \in \tilde{U}_k$ and every $d \in D$. By hypothesis 2 the solution exists and is unique on I_k, and

$$x(t) \in K \quad \forall t \in I_k. \qquad (\dagger)$$

Successor certificate. The institution computes (from its own data C_k, π_k, c_k — no observation of u^{real} needed, since $\tilde{U}_k$ is determined by the correspondence):

$$C_{k+1} := \Phi(C_k, \tilde{U}_k) \cap O^{-1}(Y_{k+1}), \quad Y_{k+1} = O(x(t_{k+1})).$$

$C_{k+1} \neq \varnothing$: the true endpoint $x(t_{k+1})$ is reachable from $x(t_k) \in C_k$ under the realized $u^{real}_k \in \tilde{U}_k$ and the realized disturbance, hence $x(t_{k+1}) \in \Phi(C_k, \tilde{U}_k)$; and $x(t_{k+1}) \in O^{-1}(Y_{k+1})$ by definition of Y_{k+1}. So Y_{k+1} is among the observation values with nonempty intersection in clause (REG)(ii), and therefore

$$(C_{k+1}, c_{k+1}) \in V. \qquad (\ddagger)$$

This closes the induction on the certificate: $(C_k, c_k) \in V$ for all $k \in \{0,\dots,N\}$ (with c_{k+1} determined at each step).

Conservative soundness (Lem2, embedded). Claim $B_k \subseteq C_k$ for all k. Base holds. Inductive step: Φ is monotone in the set argument (a union over a subset is a subset of the union over the superset) and in the action-set argument, hence

$$\begin{aligned} B_{k+1} &= \Phi(B_k, \{u^{real}_k\}) \cap O^{-1}(Y_{k+1}) \\ &\subseteq \Phi(C_k, \{u^{real}_k\}) \cap O^{-1}(Y_{k+1}) \\ &\subseteq \Phi(C_k, \tilde{U}_k) \cap O^{-1}(Y_{k+1}) = C_{k+1}. \end{aligned}$$

Conclusion. Chaining $(\dagger)$ over $k = 0,\dots,N-1$ gives $x(t) \in K$ for all $t \in [0,T]$, for every branch of observations, implementation, deployment, and disturbance, and every $x_0 \in B_0$. Causality note: C_{k+1} is computed from C_k, π_k, c_k (institution-internal) and the observation Y_{k+1} (received at t_{k+1}); by induction C_k is a function of $Y_1,\dots,Y_k$ and the initial data, so π_k is a function of the observation history — observation-based and causal. □ (Thm1 and Lem2)

Proof of R02.Prop3 (incompleteness witness)

Plant. $X = R \times \{\pm 1\}$ with coordinates (z, θ), θ a hidden mode; held commands $u \in \{-1, +1\}$ on unit intervals $I_k = [k, k+1)$; no exogenous disturbance; dynamics

$$\dot{z} = \theta u - 1.$$

Safe set $K = \{(z,\theta) : z \geq 0\}$. Read the two branches: for $\theta = +1$, $u = +1$ holds ($\dot{z} = 0$) and $u = -1$ crashes ($\dot{z} = -2$); for $\theta = -1$ the roles swap. A command safe for one mode crashes the other at rate 2; the mode-matched command always holds. Prior $C_0 = \{(3,+1), (3,-1)\}$.

Observation maps. $O^{ex}_k(z,\theta) = (z,\theta)$ for $k \geq 1$ (the exact processor receives the mode from t_1 on; at t_0 no mode information); $O^{co}_k(z,\theta) = z$ for all k (the conservative processor trusts only z — a declared coarsening, i.e. an outer processor that discards the mode channel).

Exact filter: viable forever. On $[0,1]$ the exact institution also faces the pair; the common command $u = +1$ is safe on $[0,1]$ for both branches (the $\theta=-1$ branch falls from 3 to 1, staying in K). At t_1 the mode is revealed, the compatible set becomes a singleton $\{(z,\theta)\}$, and the matched hold command keeps z constant forever: clause (REG) holds on the certificate family consisting of the initial pair and the singletons, and Thm1 certifies robust institutional viability on every horizon.

Conservative filter: genuinely nonviable. Under any z-only policy, let (z^+, z^-) be the two mode-branch positions. Whatever command is held, exactly one branch holds and the other falls at rate 2, so

$$z^+(k+1) + z^-(k+1) = z^+(k) + z^-(k) - 2 \quad \text{for every admissible command,}$$

hence $z^+ + z^- = 6 - 2k$ after k intervals. Keeping both branches in K forever requires $z^+ + z^- \geq 0$ forever — impossible: **every** z-only policy exits K by $t = 3$. Moreover the obstruction is locally visible: at t_2 the conservative compatible set (under the forced safe play of the first two intervals) is $C = \{(1,+1), (1,-1)\}$, and each member has exactly one safe command on $[2,3]$ (the mismatched command exits within time $1/2 < 1$), the two commands being opposite: the common safe-command intersection is empty, so (REG) fails at this reachable state — the Prop4 obstruction, reached by the conservative dynamics. The conservative bridge therefore correctly refuses certification, and the refusal is not an artifact: the conservative information pattern is truly nonviable.

Conclusion. Same plant, same K, same prior: the exact filter satisfies the bridge's hypotheses and the closed loop is viable; the sound conservative filter violates (REG) at a reachable state and the conservative closed loop is nonviable. Conservative assessment is sound (it never certifies an unsafe system — Lem2) but incomplete (it rejects — here, correctly at its own information level, but needlessly at the physical level — a system the exact processor governs safely): filter exactness is a semantic property of the declared assessment layer, not a refinement of the same judgment. (The witness transplants the packet's hidden-mode conflict, `sources/A001_topdown_source.txt` line 558 / Example 4.1, and Theorem 4.2's observation-empties-kernel phenomenon, into the closed-loop setting with a forced decay that makes the distinction a viability difference, not merely a certificate difference.) □

Proof of R02.Prop4 (necessity)

Suppose (C,c) is reachable, contains a boundary state, no informative observation precedes the next prescription, and the per-state safe-command intersection is empty. Any observation-based policy must select one command $u \in U$ from the information available at (C,c). Robust safety on I_k requires, in particular, safety of every compatible state $x \in C$ under that u. For each $x \in C$ the set of commands safe at x is exactly the corresponding member of the intersection; the intersection being empty, no u is safe for all $x \in C$ simultaneously. Hence every candidate first command fails, so no policy is robustly safe from (C,c), and (C,c) can belong to no certificate family on which (REG) holds (REG would demand a witness command). This is the one-step argument of corrected `06` §3, transplanted verbatim to the closed-loop setting. □

Proof of R02.Cor5

Run the induction of Thm1 conditionally on the realized measured sequence $(d^m(t_k))_k$. The witness at (C,c) may be chosen as a function $u^{cmd}(d^m)$ of the revealed value; clauses (i)–(ii) quantify over hidden components only, with the measured values fixed. All steps of the proof are unchanged, relativized to the fixed measured sequence; the conclusion holds for each such sequence separately, hence for all. □

Proof of R02.Cor6

Clause (i) verified on K_{-r} means the closed-loop velocity field at points of ∂K_{-r} satisfies the erosion-lemma inequality with total perturbation budget Δ_ε (interface-free by construction: the theorem's own realization error is the budget — observation error $a_o\ \varepsilon$ through the command map's Lipschitz constant L_κ, implementation error $a_u\ \varepsilon$, plant-model error μ, converted by $C = L_u(L_\kappa\ a_o + a_u)$ exactly as in corrected `02`). By corrected `02` Lemma 2 (clause-level match: identical hypotheses, identical conclusion), K_{-r} is strongly invariant under the closed-loop envelope, and membership $x \in K_{-r}$ implies $x(t) \in K_{-r} \subseteq K$ for all subsequent times. Substituting K_{-r} for K in clause (i) of (REG) therefore yields true-state safety in K under the stated budget condition. The feasible interval for r may be empty — this caveat is retained verbatim from corrected `02`. □

Field 9 — Counterexample showing necessity or failure outside scope

Three boundary witnesses:

1. **Common-action necessity:** Prop4 + its instantiation inside Prop3 (and the packet's own hidden-mode example, `sources/A001_topdown_source.txt` line 558, Example 4.1): individually viable compatible states with empty common safe command — information structure, not dynamics, is the failure.
2. **Latency is not cosmetic:** with $DeP \equiv \{c\}$ (full one-step latency), clause (i) must hold for the previous command on every interval; the certificate family indexed by c is what makes this checkable. In the Prop3 plant with latency one, a θ-safe command issued at t_1 deploys at t_2, so the interval $[t_1,t_2]$ runs on the parking command — safe only if parking was chosen with foresight: the certificate family must therefore distinguish c-values, which it does by construction.
3. **All-branches vs. existential:** replacing $\forall u^{real} \in I$ by $\exists$ converts the judgment into the some-branch (cherry-picking) one; the two are related by the monotonicity reversal of corrected `01` Prop 7 — enlarging I shrinks all-branches safety (Thm1) while enlarging some-branch safety. The theorem proves the governance-relevant (all-branches) direction only.

Outside scope (no claim): measurable/continuous/Lipschitz selectors for the witness correspondence (see Field 12); stochastic observation laws; RFDE history-state filters; interior events; infinite horizon without the compactness closure of R03.Lem4.

Field 10 — Interface producer/consumer contract

- **Producer object:** closed-loop guarantee `Thm1` + certificate condition `(REG)` + conservative-update soundness `Lem2`, with types: (C,c)-certificates, correspondence I/DeP, observation O, plant f.
- **Consumer objects:** (i) institutional-operationalization module (Wave 3/E1 of the research plan): must instantiate I from real authority/compliance/latency records and check (REG); (ii) Paper 5 closed-loop falsification design: the certificate condition is the falsifiable object (failure of (REG) at a reachable (C,c) is a predicted governance breakdown); (iii) R08's typed lift maps (information-state judgments of the hierarchy become instances).
- **Type/unit map:** prescriptions $u^{cmd} \in U^{cmd}$ and realized actions $u^{real} \in U$ are distinct types joined only by I (`TCS-1.0` §2.3); compatible sets carry physical units; no aggregation across blocks is performed.
- **Failure condition:** revocation if any hypothesis of Field 6 is withdrawn on an application (in particular: strategic I (CIRC-3), non-conservative filter claimed as conservative, or selector regularity assumed without proof).

Field 11 — Error, horizon, and safety erosion for approximations

Cor6 is the erosion interface: error budget $\Delta_\varepsilon = C\varepsilon + \mu$, erosion depth $r = c\varepsilon$ feasible when $(L_G\ c + C)\varepsilon + \mu \le \alpha$, $c\varepsilon < \rho$, $K_{-c\varepsilon} \neq \varnothing$; horizon finite T (infinite horizon via R03.Lem4 under compactness). The conservative filter's coarseness is not an error budget in this sense — it is exactness loss in the assessment layer, whose consequence is incompleteness (Prop3), not erosion.

Field 12 — Selector and implementation regularity

Thm1's policy is an arbitrary selector of the (REG)-witness correspondence (axiom-of-choice level, consistent with the packet's selector discipline: corrected `08` limitation 1, corrected `01` §"Measurable selection is not Lipschitz implementation"). Claiming measurable/continuous/computable π requires a selection theorem on V — an open obligation (research plan D2). Implementation regularity enters only through the declared correspondences' compactness; I may be nonconvex (all-branches quantifier is set-algebraic, no convexification needed for the judgment; convexification matters only if envelope strong-invariance tools are invoked, as in Cor6, where the relaxation-exactness caveat of corrected `03` §7 applies).

Field 13 — Stochastic/hybrid/RFDE qualifications

- **RFDE:** the proof pattern transfers to the history-space setting of corrected `08` (sampled RFDE finite-clopen knowledge kernel) with `C` a compact history set and `Φ` the exact translated-history update, provided the history-class closure conditions (uniform boundedness, equi-Lipschitz, translated-history membership — corrected `08` §RFDE history closure) hold; this qualification is declared, not proved here.
- **Hybrid:** review-synchronised events fold into the successor clause (reset maps compose with `Φ`); interior/variable events are outside scope (packet's standing gap).
- **Stochastic:** chance-constrained versions need the law-support alignment guard (schema audit QF-2; corrected `01` Prop 8); no stochastic claim is made.

Field 14 — Novelty status with exact references

- Internal: the docket T2 acceptance criteria are met; the packet's existing components (corrected `01` hierarchy; corrected `06` §2–§4; corrected `08` information-state kernel; A001 Thm 13.2 at `sources/A001_topdown_source.txt` line 1611 — which the error register lists as assumption-driven) supply the pieces, but **no packet record composes them into one closed-loop quantifier chain with the conservative/exact, latency, all-branches, and measured/hidden distinctions simultaneously**; that composition, and the sound-but-incomplete conservative-filter calculus (Lem2/Prop3), are the new content.
- External: perception/observer-based robust control and verified-controller literatures contain related compositional guarantees; **exact bibliographic matching is outstanding** (packet self-containment caveat). No external novelty claim is made.

Field 15 — Publication destination

Paper 1 (governance architecture section: the bridge as the theory's central closed-loop theorem); Paper 2 (theorem atlas: Thm1 + Lem2 + Prop3 with full proof); Paper 5 (closed-loop empirical design: (REG) as the falsifiable certificate). Monograph: chapter on the execution chain.

Field 16 — Remaining obligations and revocation triggers

Obligations: selector regularity (D2); RFDE/history-space instantiation with the closure conditions verified on one application; stochastic variant with support alignment; external novelty audit (F1); an instantiated institutional case (E1/E2). Revocation triggers: withdrawal of the non-strategic `I` declaration (CIRC-3) — the theorem's quantifier chain would become a fixed-point claim; discovery that a conservative update is not inclusion-monotone (breaks Lem2); any application asserting (REG) without exhibiting the certificate family.

Field 17 — Machine-readable dependency edges


```json
{
  "result_id": "R02",
  "target": "T2",
  "depends_on": [
    "corrected_theorems/01_operator_I_hierarchy.md (judgment quantifiers, Prop 7)",
    "corrected_theorems/06_A001_selected_operatorI_audit.md (common-action obstruction §3, delayed-information §4)",
    "corrected_theorems/08_A002_sampled_hybrid_audit.md (sampled kernel chain, selector caution)",
    "corrected_theorems/02_operator_I_strong_invariance_and_erosion.md (Lemma 2, for Cor6)"
  ],
```

```
  "unblocks": ["T8 typed hierarchy maps (R08)", "institutional
  operationalization (Wave E)", "Paper 5 closed-loop design", "R04
  policy/information map field"],
  "status": {"R02.Thm1": "proved", "R02.Lem2": "proved", "R02.Prop3": "proved",
  "R02.Prop4": "proved", "R02.Cor5": "proved", "R02.Cor6": "proved (clause-level
  assembly)"},
  "mapping_type": "EXACT_SPECIALIZATION (of the TCS-1.0 execution chain) +
  COUNTEREXAMPLE_OR_LIMIT (Prop3)",
  "novelty": "internal-new composition; external check outstanding"
}
```

Result Record R03 — Docket T3: The Viability–Diagnostic Bridge

Field 1 — Result ID and target docket item

R03 (R03.Thm1 certificate trichotomy; R03.Thm2 stock-to-rate margin failure; R03.Thm3 aggregate margins are not kernels; R03.Lem4 compactness horizon closure; R03.Cor5 combined-error erosion conversion). Target: **T3** ("determine when a diagnostic margin or finite-horizon estimate is sound for nonviability; complete for viability; an inner/outer certificate; merely descriptive").

Field 2 — Verdict

Repairable → proved (assembled + new witnesses). The packet holds all positive components (adversarial exit, erosion lemma, local-horizon bracket) but no record states the bridge classification or the necessity counterexamples; this record assembles the positives with clause-level matches and supplies the new negative witnesses the docket demands ("counterexample showing why local stock-to-rate or aggregate margins are not kernels").

Field 3 — Exact statement

Setting. Diagnostics are functionals `M(x)` computed from model + observation data at a state (e.g., robust finite-horizon value `V_T(x)` in the sense of A001 Definition 15.1 at `sources/A001_topdown_source.txt` lines 1796–1804; distance-type margins `dist(x, ∂K)`; stock-to-rate ratios `(S − S_min)/r`; aggregate margins `dist(P(x), ∂P(K))`).

R03.Thm1 (certificate trichotomy). With `RViab_T(K)` the robust viability kernel on horizon `T` and `RViab_∞(K)` its infinite-horizon limit:

1. **(Outer / sound for nonviability.)** If `M` is an adversarial-exit certificate — there exist `q, a, ε > 0`, a strip `S_a = {0 ≤ q ≤ a}`, and the lower-game information pattern such that for every admissible control and every state in the strip an admissible nonanticipative disturbance realizes `D^+q ≤ −ε`, with existence up to exit — then `M(x) < 0` certifies `x ∉ RViab_∞(K)`, and the exit occurs within time `a/ε`. The certificate is sound for nonviability and requires no policy search.
2. **(Inner / sound for viability.)** If `M` is a margin-with-budget certificate — `K` satisfies the erosion-lemma hypotheses (two-sided tubular radius `ρ`, `C^{1,1}` signed distance, normal correspondence), the certified closed-loop envelope has Hausdorff-Lipschitz gain `L_G` and boundary margin `α`, and the diagnostic error budget satisfies `Δ = Δ_ε + μ` with `L_G r + Δ ≤ α`, `0 < r < ρ`, `K_{−r} ≠ ∅` — then `M(x) ≥ r` certifies `x ∈ RViab_∞(K_{−r}) ⊆ RViab_∞(K)`.
3. **(Descriptive.)** If `M` carries neither an adversarial quantifier pattern (case 1) nor a margin/budget/regularity package (case 2), then `M` implies neither nonviability nor viability: both failure directions are witnessed (Thm2: soundness failure of the safety direction for stock-to-rate margins; Thm3: kernel-hood failure for aggregate margins; and the packet's own soft-landing failure for the nonviability direction, see Field 9).

R03.Thm2 (stock-to-rate margins are unsound without a rate-persistence band). For the stock process `Ẋ = −e^t` with `K = {X ≥ 0}` and `S_min = 0`: the stock-to-rate diagnostic horizon from `X(0) = a` is `T_diag = a/r(0) = a`, while the true exit time is `T* = ln(1 + a)`. The ratio `T_diag/T* = a/ln(1+a) → ∞` as `a → ∞`: the overestimate is unbounded, so no constant-factor correction can restore soundness. The rate-band hypothesis of the A002 local-horizon bracket (`thm:horizon`, `sources/A002_general_theory_source.txt` line 2161; accepted conditionally in corrected `09`) is therefore not merely technical: without it, stock-to-rate margins are descriptive only.

R03.Thm3 (aggregate margins are not kernels). In the system of R01.Thm2 (`ẋ_1 = c+u, ẋ_2 = −(c+u), c > 1, S = [0,1]²`, aggregate `P = x_1+x_2`), the aggregate margin `dist(P(x), ∂P(S))` is constant and positive along **every** trajectory (`P ≡ const ∈ (0,2)` for

$x \in \mathrm{int}\ S$), for all time; yet $\mathrm{RViab}(S) = \mathrm{CViab}(S) = \varnothing$. Aggregate margins are therefore not viability kernels in either direction, and no erosion of the aggregate level set restores safety (the erosion direction is orthogonal to the failing components).

R03.Lem4 (compactness closure of the horizon limit). For the sampled robust problem with compact state space (or a compact invariant enclosure), compact action sets, and successor maps Succ(x,u,d) that are Hausdorff-upper-semicontinuous in u uniformly in (x,d) with closed values: defining $R_0 = K$ and $R_{n+1} = \{x \in K : \exists u\ \forall d : \mathrm{Succ}(x,u,d) \subseteq R_n\}$, the family (R_n) is decreasing and

$$R_\infty := \bigcap_n R_n = \mathrm{RViab}_\infty(K):$$

a state is robustly viable on every finite horizon if and only if it is robustly viable on the infinite horizon, and R_∞ is the largest robustly invariant subset of K. Consequently the hierarchy's non-implication 8 (corrected 01: "Finite-horizon viability does not imply infinite-horizon viability without a closure/compactness argument") closes exactly at these hypotheses: without compactness/upper semicontinuity the implication fails; with them, viability at all finite horizons is equivalent to infinite-horizon viability.

R03.Cor5 (combined-error erosion conversion). If a margin certificate is computed with observation error $\le a_o\ \varepsilon$, implementation error $\le a_u\ \varepsilon$, and plant-model error $\le \mu$, then under the erosion-lemma hypotheses the certified inner kernel is K_{-r} with

$$(L_G\ c + C)\varepsilon + \mu \le \alpha, \quad C = L_u(L_\kappa\ a_o + a_u), \quad r = c\varepsilon < \rho,$$

exactly the error conversion of corrected 02; the feasible interval for c may be empty, in which case the diagnostic certifies nothing.

Field 4 — State and phase space

Thm2: scalar ODE, $X = R_{\ge 0}$, nonautonomous rate (declared model class: ODE with known time-varying rate — the diagnostic's information). Thm3: R^2 with noncompensatory box constraint and aggregate projection (as R01.Thm2). Lem4: sampled system, compact X, U, D, successor correspondence (the packet's canonical sampled setting, corrected 08).

Field 5 — Quantifier order and information pattern

Thm1(1): $\forall\pi\ \exists$ nonanticipative disturbance : exit by a/ε (adversarial lower game; corrected 06 §7's repaired quantifiers). Thm1(2): $\exists\kappa\ \forall d\ \forall\varphi$: stay (all-solutions strong invariance; corrected 02 Theorem 1 pattern). Thm2/Thm3: no quantifier pattern — that is the point: the diagnostics carry none, and the theorems show the consequences. Lem4: $\exists u\ \forall d$ per step, $\exists\pi\ \forall d\ \forall\varphi$ globally (robust kernel).

Field 6 — Assumptions, including existence/completeness

Thm1(1): measurable/nonanticipative disturbance selection and existence up to exit (as in corrected 06 §7). Thm1(2)/Cor5: erosion-lemma regularity (corrected 02 Lemma 2 hypotheses, including the normal correspondence and the exclusion of arbitrary closed sets — the packet's $K = \bigcup_j [2j, 2j+1]$ counterexample for uniform erosion remains controlling). Lem4: compactness + Hausdorff-usc successor maps (explicitly the hypotheses the packet's sampled kernel theorems already carry: corrected 08 assumes Hausdorff continuity, which is stronger than the upper semicontinuity used here). Thm2/Thm3: Lipschitz dynamics, unique forward-complete solutions.

Field 7 — Mapping type

Thm1: EXACT_SPECIALIZATION of the classification question onto the packet's two existing certificate types + COUNTEREXAMPLE_OR_LIMIT for the residual class. Thm2: COUNTEREXAMPLE_OR_LIMIT (necessity witness for thm:horizon's hypothesis). Thm3: COUNTEREXAMPLE_OR_LIMIT (reuses R01.Thm2's system — internal dependency edge). Lem4: EXACT_SPECIALIZATION of classical viability-limit machinery to the packet's sampled class, with proof supplied for self-containment. Cor5: clause-level theorem match to corrected `02`.

Field 8 — Self-contained proof

Proof of R03.Thm1

(1) is the corrected adversarial-exit theorem of A001 Theorem 5.2 (repaired quantifiers and conclusion; corrected `06` §7 accepts it): for every control policy there exists a nonanticipative disturbance strategy driving `q` across 0 within `a/ε`; hence `x ∉ RViab_T(K)` for `T ≥ a/ε`, and a fortiori `x ∉ RViab_∞(K)`. Clause-level match; no new mathematics.

(2) is corrected `02` Lemma 2 with its error conversion: the margin `α` at `∂K`, the Lipschitz budget `L_G r`, and the total perturbation `Δ` give strong invariance of `K_{−r}` under the certified closed-loop envelope; membership `x ∈ K_{−r}` therefore implies eternal stay in `K_{−r} ⊆ K` under all declared disturbances and all solutions (all-solutions semantics via the envelope inclusion, corrected `02` Theorem 1). Clause-level match.

(3) is established by the two witnesses below plus the packet's soft-landing witness: a diagnostic with neither the adversarial pattern nor the margin/budget package can point in either direction and be wrong — soundness fails in the safety direction (Thm2: `T_diag` overestimates exit time unboundedly; Thm3: aggregate margin positive forever while the kernel is empty) and completeness fails in the nonviability direction (Field 9: soft landing never exits while `T_diag` is finite). □

Proof of R03.Thm2

Rate `r(t) = e^t`; from `X(0) = a > 0`: `X(t) = a − (e^t − 1)`, so `X` reaches 0 at `T* = ln(1+a)` exactly. The diagnostic at `t = 0` uses the current rate `r(0) = 1`, giving `T_diag = X(0)/r(0) = a`. Ratio `a/ln(1+a)`: at `a = 10` it is `10/ln 11 ≈ 4.17`; as `a → ∞` it diverges. For every claimed soundness factor `F` there is an initial stock with `T_diag > F·T*`: the diagnostic asserts safety for a horizon that exceeds the true horizon by more than any fixed factor. Since the A002 local-horizon bracket's hypothesis ("rate remains within the declared relative band on an interval already long enough to force crossing" — corrected `09`, adjudication row "Local-horizon bracket") is violated here by construction (`r` leaves every band `[r(0)(1−δ), r(0)(1+δ)]` exponentially fast), the witness proves that hypothesis necessary. □

Proof of R03.Thm3

Immediate from R01.Thm2 (proved there): `P(x(t)) ≡ P(x(0)) ∈ P(S)` for every trajectory, so the aggregate margin `dist(P(x(t)), ∂P(S))` is a positive constant along every branch for all time — the aggregate diagnostic can never register any deterioration. Meanwhile `RViab(S) = CViab(S) = ∅` (R01.Thm2 Step 2: both components are driven out within time `1/(c−1)` under every control). Erosion of aggregate level sets `P^{-1}([δ, 2−δ])` intersects `S` in strict subsets but every such subset still has empty kernel, since the destruction mechanism (fibre-internal transfer at rate `≥ c−1 > 0`) is invisible to `P` and unaffected by `δ`. □

Proof of R03.Lem4

Write `Pre(W) := {x ∈ K : ∃u ∀d : Succ(x,u,d) ⊆ W}` so `R_{n+1} = R_n ∩ Pre(R_n) = Pre(R_n)` (`Pre(W) ⊆ K` by construction and `R_n ⊆ K`).

Monotonicity. `R_1 = Pre(K) ⊆ K = R_0`; if `R_n ⊆ R_{n−1}` then `Pre(R_n) ⊆ Pre(R_{n−1})`, so `R_{n+1} ⊆ R_n`: decreasing. Each `R_n` is compact (closed subsets of the compact enclosure; closedness of `Pre` follows from closed values of `Succ` and compactness of `U, D` — standard, and the same closedness the packet's sampled kernel theorem uses).

Set algebra (no continuity needed). For any decreasing family `W_n ↓ W_∞` with closed values:

`Pre(W_∞) = ⋂_n Pre(W_n).`

Indeed, $x \in Pre(W_\infty) \Leftrightarrow Succ(x,u,d) \subseteq W_\infty \subseteq W_n$ for some u and all $d \Leftrightarrow x \in \bigcap_n Pre(W_n)$; conversely $x \in \bigcap_n Pre(W_n)$ gives, for some fixed u — careful: the witness u may depend on n; the fix is the next step.

Fixed point. Claim $R_\infty = \bigcap_n R_n$ satisfies $R_\infty = Pre(R_\infty)$; in particular R_∞ is robustly invariant (witness command at every point), and it is the largest such subset of K (any robustly invariant $W \subseteq K$ satisfies $W \subseteq Pre(W) \subseteq \dots \subseteq R_n$ for all n).

Inclusion *$Pre(R_\infty) \subseteq R_\infty$*: $Pre(R_\infty) \subseteq Pre(R_n) = R_{n+1} \subseteq R_n$ for every n; intersect.

Inclusion *$R_\infty \subseteq Pre(R_\infty)$*: let $x \in R_\infty$. For each n, $x \in R_{n+1} = Pre(R_n)$, so there is $u_n \in U$ with $Succ(x,u_n,d) \subseteq R_n$ for all d. By compactness of U, a subsequence $u_{n_j} \to u^* \in U$. Fix m; for $n_j \geq m$, $Succ(x, u_{n_j}, d) \subseteq R_{n_j} \subseteq R_m$. By Hausdorff-upper-semicontinuity of $Succ(x,\cdot,d)$ at u^* and closedness of R_m: $Succ(x, u^*, d) \subseteq R_m$. This holds for every m, hence $Succ(x,u^*,d) \subseteq \bigcap_m R_m = R_\infty$ for all d, i.e. $x \in Pre(R_\infty)$.

Equivalence with infinite-horizon viability. R_∞ robustly invariant and $\subseteq K \Rightarrow$ every $x \in R_\infty$ is robustly viable forever (recursively choose the witness command; concatenation is the policy — an arbitrary selector, Field 12). Conversely, if x is robustly viable on every finite horizon n, then by backward induction $x \in R_n$ for every n (the standard finite-horizon characterization the packet's sampled kernel theorem supplies; corrected 08), so $x \in R_\infty$. Hence $R_\infty = RViab_\infty(K)$, and finite-horizon viability at all horizons is equivalent to infinite-horizon viability. □

Proof of R03.Cor5

Clause-level assembly of corrected 02 (Lemma 2 + error conversion): the three error sources enter the perturbation budget of the closed-loop envelope as $\Delta_\varepsilon = C\varepsilon$ with $C = L_u(L_\kappa a_o + a_u)$ plus the model error μ (added to the budget since the envelope inclusion $\bar{G}_\varepsilon \subseteq G + \Delta B$ carries it linearly); the erosion condition $L_G r + \Delta_\varepsilon + \mu \leq \alpha$ is then exactly the lemma's hypothesis, and the conclusion is strong invariance of K_{-r}, $r = c\varepsilon$. Empty-feasible-interval caveat retained. □

Field 9 — Counterexample showing necessity or failure outside scope

- **Thm2's reverse direction (completeness failure):** the soft-landing process $\dot{X} = -cX^2$ never reaches 0 ($X(t) = X(0)/(1+cX(0)t) \to 0$ only asymptotically), so the true horizon is ∞ while the stock-to-rate diagnostic gives the finite value $X(0)/(cX(0)^2) = 1/(cX(0))$: margins cannot certify safety either — both directions of the "descriptive" verdict are witnessed.
- **Lem4's necessity:** dropping compactness, the standard escape-to-infinity failure applies (the packet's own remark on A001 Theorem 14.2: "Without dissipativity, mass can escape to infinity and the nested-empty-intersection argument fails" — the same mechanism blocks the horizon limit without confinement); dropping Hausdorff-usc, the witness subsequence argument fails and R_∞ can fail to be a fixed point (predecessor chains can shrink strictly in the limit — the outer-semicontinuity pathology the packet records in A002's Proposition on OSC insufficiency, corrected 08).
- **Erosion outside regular sets:** the packet's $K = \bigcup_{j\geq1}[2j,2j+1]$ counterexample (corrected 02) remains controlling for Thm1(2): without tubular regularity, margins do not convert to invariant erosions at any constant factor.

Field 10 — Interface producer/consumer contract

- **Producer:** the trichotomy classification + Lem4 + Cor5, typed over (diagnostic functional, certificate pattern, error budget, horizon).
- **Consumers:** Paper 5 falsification design (a proposed empirical diagnostic must declare which trichotomy class it is in; only class (1) may be cited as a falsifier of viability claims, only class (2) as a safety certificate); Paper 3 componentwise diagnostics (Thm3 prohibits aggregate-level depletion claims — ties to the componentwise deficit

design $\Lambda = [D]_+$ of corrected 05); R04's admission certificate (diagnostic field must instantiate Cor5's budget).

- **Failure condition:** revocation if the adversarial-exit theorem or erosion lemma hypotheses are weakened on an application; any aggregate diagnostic used as a kernel claim violates Thm3 and must be rejected at review.

Field 11 — Error, horizon, and safety erosion for approximations

Cor5 is the complete statement: error $(a_o, a_u, \mu) \to \Delta_\varepsilon = C\varepsilon + \mu$, horizon ∞ (via Lem4 under compactness; finite otherwise), erosion $r = c\varepsilon$ with $(L_G c + C)\varepsilon + \mu \le \alpha$, $c\varepsilon < \rho$, $K_{-c\varepsilon} \neq \varnothing$; feasible interval possibly empty. Thm2 makes precise why rate diagnostics have no constant conversion factor: the error is state/path-dependent and unbounded.

Field 12 — Selector and implementation regularity

Lem4's policy is an arbitrary selector of the witness correspondence (consistent with the packet's selector discipline); measurable/continuous selector claims need a selection theorem (open obligation D2, shared with R02). Thm1(2) requires the implementable feedback's regularity exactly as corrected 02 states (Lipschitz κ or re-checked Filippov envelope) — a measurable selector alone does not establish the erosion conversion.

Field 13 — Stochastic/hybrid/RFDE qualifications

Thm1(1): the adversarial pattern is the deterministic lower game; a stochastic analogue requires the law-support alignment (QF-2) and becomes a chance-nonviability statement — not claimed. Lem4: sampled class only; the RFDE analogue needs the history-space compactness (equi-Lipschitz history classes, corrected 08) — declared, not proved. Hybrid: interior events void the predecessor algebra (event-time branching); review-synchronised events compose as in corrected 08.

Field 14 — Novelty status with exact references

Internal: Thm1's trichotomy is the missing classification the docket demands; Thm2 and Thm3 are new witnesses; Lem4 supplies the compactness closure that corrected 01's non-implication 8 explicitly leaves open. External: the finite-to-infinite horizon closure is classical viability/dynamic-programming machinery (the packet itself cites the viability literature for kernel limits, A002 line 69: `aubin1991`, `saintpierre1994`); the trichotomy packaging and the unbounded-ratio stock-to-rate witness are, to internal knowledge, new — **external literature check outstanding**, no bibliographic novelty claim made.

Field 15 — Publication destination

Paper 2 (theorem atlas: Lem4 + Thm1 as the diagnostic-bridge section); Paper 5 (core methodological content: the trichotomy as the falsification-design grammar, Thm2/Thm3 as boxed counterexamples); Paper 3 (Thm3's prohibition on aggregate depletion claims, cited from the componentwise diagnostics section).

Field 16 — Remaining obligations and revocation triggers

Obligations: instantiate Cor5's constants on the selected empirical case (E1); external novelty check; verify Hausdorff-usc of the application successor maps (Lem4's hypothesis — the packet's theorems assume the stronger Hausdorff continuity, so applications verified under corrected 08 inherit it). Revocation triggers: any use of a stock-to-rate or aggregate margin as a kernel/certificate claim (violates Thm2/Thm3 — reviewer-enforceable); withdrawal of compactness in an infinite-horizon application of Lem4.

Field 17 — Machine-readable dependency edges


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    "corrected_theorems/06_A001_selected_operatorI_audit.md (§7 adversarial
    exit)",
    "corrected_theorems/09_A002_reduction_diagnostic_audit.md (local-horizon
    bracket scope)",
    "corrected_theorems/08_A002_sampled_hybrid_audit.md (sampled kernel
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Result Record R04 — Docket T4: Domain Admission and Projectability Certificate

Field 1 — Result ID and target docket item

`R04` (R04.Thm1 exact-admission certificate; R04.Cor2 approximate admission; R04.Tab3 classification of the included domain modules). Target: **T4** ("state necessary and sufficient conditions for exact admission; otherwise provide an approximation theorem … theorem/certificate that can classify groundwater, phosphorus, or fisheries models without verbal analogy").

Field 2 — Verdict

Proved at the structural level. The certificate theorem is proved with both directions; the classification table instantiates it on the three included domain families (groundwater A005, phosphorus A004, fisheries A001/A012/A018/A014) using only their source-declared objects, with the open items flagged exactly as the live error register records them. Empirical calibration is explicitly out of scope (packet README, excluded priorities).

Field 3 — Exact statement

R04.Thm1 (exact admission certificate)

Let `M` be a domain model with declared objects: state space `X_M`, solution concept `S_M` (with disturbance class `D_M` and admissible control/action correspondence `U_M`), constraint registry generating safe set `K_M ⊆ X_M`, and judgment family `J_M` (any of the eight `TCS-1.0` §4 judgments). Let `A_q` be a `TCS-1.0` architecture realization. Say **`M` is exactly admitted into `A_q`** if there exist:

1. **type/unit map** `τ`: a bijection between the declared stocks/fluxes/units of `M` and typed blocks of `Z_q` (moieties preserved; donor/recipient roles preserved);
2. **phase-space map** `φ`: `X_M → Z_q`, injective, with `φ^{-1}` defined on `φ(X_M)` and both maps respecting the declared block structure;
3. **dynamics correspondence**: for every `(x, u, d)`, the solution sets correspond: `φ(Sol_{S_M}(x, u, d)) = Sol_{S_q}(φ(x), τ_U(u), τ_D(d)) ∩ φ(X_M)`, and the disturbance/control classes correspond under `τ_U, τ_D`;
4. **safe-set correspondence**: `φ(K_M) = K_{q,Ω} ∩ φ(X_M)`;
5. **policy/information correspondence**: the causal information structures correspond (every `M`-policy lifts to a `P_q`-policy with identical information, and conversely on `φ(X_M)`).

Then every canonical judgment transfers exactly: for every *`J`* in the judgment family and every compatible horizon,

`x ∈ J_M(M)  ⇔  φ(x) ∈ J(A_q),`

and the kernels correspond: *`φ(J-kernel of M) = (J-kernel of A_q) ∩ φ(X_M)`*. Conversely, if the phase-space map or the safe-set correspondence or the solution correspondence fails, there exist instantiations of the same signature in which kernel equality fails (Fields 8, 9), so no judgment transfer is possible without the certificate; the failure mode identifies which map is missing.

R04.Cor2 (approximate admission)

If the dynamics correspond only up to defect `‖f_M − f_q∘φ‖ ≤ ε` on a compact domain, all other maps exact, then on horizon `T` the trajectories deviate by at most `(ε/L)(e^{LT} − 1)` (Grönwall, `L` the joint Lipschitz constant), and kernel claims transfer only through the erosion conversion of R03.Cor5 (`K_{−r}`, `(L_G c + C)ε ≤ α`) — approximate admission is an `APPROXIMATION` mapping with explicit error, horizon, and safety erosion, never an `EXACT_SPECIALIZATION`.

R04.Tab3 (classification; summary — full table in Field 8)

Module	Phase-space map	Model class / mapping type	Blocking items (live error register)	Admission verdict
Groundwater (A005)	identity on `(H_f, H_s, M_q, σ_sal, χ)`; storage via constitutive `A_i = A_i(H_i)` (DAE block)	sampled_hybrid + DAE; `EXACT_SPECIALIZATION` once open fields close	B.5 items: `V-A005-04` (route/remove `q_rel`), `V-A005-05` (leakage limiter), `V-A005-06` (total storage + jump identities), `V-A005-07` (donor/recipient positivity), `V-A005-11` (compatible-state topology)	**conditionally admissible —** certificate fields (1),(3),(5) completable after B.5; reduction to head-space dynamics is constitutive, not a `PROJECTABLE_REDUCTION` claim
Phosphorus (A004)	identity on soil-P pools	hybrid (ODE + jump balance); `EXACT_SPECIALIZATION` once open fields close	B.4 items: `V-A004-03` (hybrid jump balance), `V-A004-05` (define χ, functional dynamics), `V-A004-06` (upper soil-P bound), `V-A004-08` (trade routing/delay/conservation), `V-A004-09` (compatible-state topology)	**conditionally admissible —** same structure as A005; the jump balance is the decisive missing object for map (3)
Fisheries — resource–sink (A001 §§6–10)	identity on `(S, K)` blocks	ODE; `EXACT_SPECIALIZATION` — **complete** (no blocking items: the A001 corrections B.1 are proof-wording repairs, not missing objects)	none for admission	**admitted**

Module	Phase-space map	Model class / mapping type	Blocking items (live error register)	Admission verdict
Fisheries — C3/C4 RFDE (A012/A018)	`(N_t, Z_t, E_t)` resp. `(N_t, A_t, Z_t, E_t)` history spaces	RFDE; `EMBEDDING` into `C([−τ,0], R^n)` with corrected `05`'s phase-state declarations and model-version tags (`NV-012`, `NV-013`)	version discipline only (gated/ungated, working/QSS)	**admitted at exact model-version scope**
Fisheries — cod (A014)	scalar phase line	ODE; `EXACT_SPECIALIZATION` of the corrected scalar-autonomous obstruction (error register B.11)	B.11 corrections (nonautonomous `C(t)`, threshold-trichotomy demotion)	**admitted at corrected status**

Field 4 — State and phase space

Per module as in Tab3 — the theorem is precisely the discipline that forces the phase-space declaration: an admission record is incomplete until `φ`, `X_M`, `Z_q` and the solution concepts are named (A005 declares `Z = (H_f, H_s, M_q, σ_sal, χ)` with `A_i = A_i(H_i)`; A004's pool vector; A001's `(S,K)`; A018's history states per corrected `05`).

Field 5 — Quantifier order and information pattern

The certificate requires judgment-quantizer alignment: the `∃π ∀w ∀φ` chains of `M` and `A_q` must correspond under the lift of map (5); measured/hidden disturbance splits and observation fibres must be carried by `τ_D` and the information correspondence. A domain module whose policy class observes more than the canonical chain permits is not admitted without enlarging the architecture's `O_q/P_q` declarations (the `A005/A004` compatible-state items are exactly this requirement).

Field 6 — Assumptions, including existence/completeness

Maps (1)–(5) exist (that is the certificate); solution concepts well-posed on both sides (existence/uniqueness or the set-valued declared concept); `K_M` closed; horizons compatible. For Cor2: compact domain, joint Lipschitz `L`, defect `ε` uniform on the domain.

Field 7 — Mapping type

The certificate is the machine that decides mapping types: exact admission → `EXACT_SPECIALIZATION` or `EMBEDDING` (per `φ`'s codomain); defective dynamics → `APPROXIMATION` (Cor2); claimed-but-failed reduction → `REJECTED_MAPPING` (Field 9); verbal-only similarity → `ANALOGY_ONLY` (excluded from transfer by the theorem).

Field 8 — Self-contained proof

Proof of R04.Thm1 (sufficiency)

Fix a horizon and a judgment `J` with quantifier pattern `∃π ∀w ∀φ` (robust chain; the other patterns are identical with the obvious quantifier relabeling).

Trajectory correspondence. By map (3), for every x, π-lift, w: φ maps S_M-solutions from (x, u, w) onto the S_q-solutions from $(\varphi(x), \tau\_U(u), \tau\_D(w))$ that remain in $\varphi(X\_M)$. By map (5), the policy classes correspond: every M-policy π_M lifts to π_q with $\pi\_q(\varphi\text{-history}) = \tau\_U(\pi\_M(\text{history}))$ and conversely. Consequently the closed-loop solution sets correspond exactly:

$$\varphi(Sol\_\{S\_M\}(x, \pi\_M, w)) = Sol\_\{S\_q\}(\varphi(x), \pi\_q, \tau\_D(w)).$$

Safe-set correspondence. Map (4): a trajectory stays in K_M iff its image stays in $K\_\{q,\Omega\}$ (on $\varphi(X\_M)$).

Judgment transfer. The M-statement "$\exists\pi\_M\ \forall w\ \forall\varphi\_M\!: \varphi\_M([0,T]) \subseteq K\_M$" translates, under the two correspondences above, symbol-by-symbol into the A_q-statement with π_q, $\tau\_D(w)$, and $K\_\{q,\Omega\}$: same quantifiers, corresponding domains. Hence truth values coincide at $x \leftrightarrow \varphi(x)$, and the kernel (the set of states where the statement holds) corresponds. The argument is uniform over the judgment family because it never touches judgment-specific structure beyond the quantifier pattern. □

Proof of R04.Thm1 (necessity direction, by failure-mode witnesses)

- **Solution correspondence fails** (same state space, different dynamics): $X = R$, $K = [0,1]$, model A: $\dot{x} = u$, $u \in [-1,1]$; model B: $\dot{x} = u + 1$. With $\varphi = id$ and identical constraint/control declarations, the viability kernel of A contains $1/2$ (hold $u = 0$), while B's kernel within K is empty (persistent drift +1: every trajectory exits in finite time; at $x = 1$ exit is immediate under all controls). Same types, same units, same safe set — no judgment transfer: map (3) is indispensable.
- **Safe-set correspondence fails:** two models differing only by one additional component constraint (e.g., an upper soil-P bound, V-A004-06): kernels differ by exactly that constraint's erosion; transfer without map (4) silently imports the extra constraint (or drops it) — both directions fail.
- **Policy/information correspondence fails:** the common-action obstruction (R02.Prop4 / A001 Example 4.1): a domain policy class that observes a hidden mode has a nonempty kernel where the canonical chain without that observation has an empty one — no transfer without map (5).

Hence each map's absence is witnessed by a pair of same-signature instantiations with different kernels: the certificate is necessary in the strong sense that every missing field is independently load-bearing. □

Proof of R04.Cor2

Standard Grönwall comparison on the compact domain: two solutions with vector fields differing by $\leq \varepsilon$ and common Lipschitz L deviate by $\|x\_M(t) - \varphi^\{-1\}(x\_q(t))\| \leq (\varepsilon/L)(e^\{Lt\} - 1)$ on $[0,T]$. A trajectory certified to keep distance $\geq r$ from ∂K with $(L\_G\ c + C)\varepsilon \leq \alpha$ (R03.Cor5 budget with trajectory error in place of the perturbation budget) keeps the true state in K. The mapping is therefore APPROXIMATION with error $(\varepsilon/L)(e^\{LT\}-1)$, horizon T, erosion r — and never exact. □

Classification evidence (Tab3 justification)

- **A005 groundwater** (sources/full/A005_Paper_III_Groundwater_Module.txt): declares boundary/units (§2), state $Z = (H\_f, H\_s, M\_q, \sigma\_sal, \chi)$, constitutive storage $A\_i = A\_i(H\_i)$, $C\_i = dA\_i/dH\_i > 0$ (a DAE block per TCS-1.0 §2.2), sampled pumping/recharge controls, compatible-state uncertainty, and explicitly frames itself as "an admissible model template … not yet a calibrated case study" — exactly the admission-certificate posture. The open error-register items B.5 (V-A005-04/05/06/07/11) each block one certificate field: storage/jump identities block map (4)'s closure; compatible-state topology blocks map (5). Verdict: conditionally admissible; the conditions are enumerable, not verbal.

- **A004 phosphorus** (`sources/full/A004_Paper_IV_Phosphorus_Module.txt`): pool vector and ledger structure declared; the missing hybrid jump balance (`V-A004-03`) blocks map (3) at event branches; undefined χ/functional dynamics (`V-A004-05`) block map (2)'s block structure; trade routing and compatible-state topology (`V-A004-08/09`) block maps (3)/(5).
- **A001 §§6–10 fisheries**: closed equations, declared constraint sets, proven kernels (Theorems 6.2–6.5, 10.2) — all five maps are identities or explicit correspondences already present in the source; admitted.
- **A012/A018 C3/C4**: history-state phase spaces and model-version discipline are already fixed by corrected `05` and the notation registry (`NV-012/013`); admitted at tagged version scope (the `DYN-C4-WORKING` vs `DYN-C4-QSS` separation is enforced, never merged).
- **A014 cod**: admitted at the corrected scalar-autonomous status (error register B.11: the sound exact result is the phase-line obstruction; the finite-time threshold trichotomy is demoted).

Field 9 — Counterexample showing necessity or failure outside scope

The three failure-mode witnesses inside the necessity proof (dynamics mismatch; safe-set mismatch; information mismatch) are the counterexamples. Additional scope note: the A018 seam (corrected `05`) is the standing example of a rejected dynamic admission (`LEDGER-PRIM-CLOSED-v1 ↛ DYN-C4-WORKING`, `REJECTED_MAPPING` with reasons 1–5 recorded there): the present theorem generalizes that discipline from one seam to all domain admissions.

Field 10 — Interface producer/consumer contract

- **Producer:** the admission certificate (five maps + verdict + blocking list) as a per-module record.
- **Consumers:** the empirical case selection (E1: the readiness comparison becomes "which module closes its blocking list soonest"); Papers 3–5 (each domain instantiation must cite its admission record, not verbal similarity); the concordance (`01_canonical_concordance_A001_A025.csv` gains an admission-verdict column per domain row).
- **Failure condition:** any cross-module theorem transfer citing a module whose blocking list is nonempty without an `APPROXIMATION` row carrying Cor2's triple — reviewer-enforceable rejection.

Field 11 — Error, horizon, and safety erosion for approximations

Cor2: error `(ε/L)(e^{LT}−1)`; horizon `T` explicit (exponential growth in `T` — no uniform-in-time claim); safety erosion `r = cε` via R03.Cor5 with the feasible-interval caveat. Exact admission carries no error and no horizon restriction beyond the judgment's own.

Field 12 — Selector and implementation regularity

The certificate is selector-agnostic (judgment transfer is at the policy-class level); implementation regularity enters only through map (5)'s correspondence of implementation branches — an `I_q` declaration must exist for modules claiming institutional judgments (A004/A005 do not yet declare one — flagged in their blocking lists implicitly through map (5)).

Field 13 — Stochastic/hybrid/RFDE qualifications

Hybrid modules (A004, A005) require map (3) to include event branches (jump balances — hence the decisive role of `V-A004-03`); RFDE modules (A012/A018) require history-space φ (identity on histories; the translated-history closure conditions of corrected `08` apply to any filter claim); stochastic domain laws would require the QF-2 support declaration — none of the three included modules declares one, so no chance-level admission is claimed.

Field 14 — Novelty status with exact references

Internal: the docket demands exactly this certificate ("type/unit map; phase-space map; projectability; policy/information map; error bound; horizon; safety erosion; artifact/version identity"); no packet record states the necessity direction with per-map witnesses; corrected `05` is the single-instance precedent. External: admission/classification certificates of this form are standard practice in formal modelling (model-class embeddings, morphism-based specification frameworks); **the specific five-map formulation tied to the TCS judgment family is, to internal knowledge, new packaging; external literature check outstanding**; no bibliographic claim made.

Field 15 — Publication destination

Paper 1 (admission standards section — the certificate as the theory's domain gate); Paper 2 (Thm1's short proof in the mapping-types section); monograph (the classification table as the standing domain appendix).

Field 16 — Remaining obligations and revocation triggers

Obligations: close the A004/A005 blocking lists (the error-register actions already registered); instantiate the certificate rows in the concordance; select and execute the empirical case (E1). Revocation triggers: any silent change of a module's declared objects (version identity axiom 7); a claimed admission whose map (3) omits event branches or history structure.

Field 17 — Machine-readable dependency edges


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Result Record R05 — Docket T5: Tubular Assume–Guarantee Composition with a Gain Operator and Feasibility Fixed Point

Field 1 — Result ID and target docket item

R05 (R05.Thm1 contract-amplitude composition [Version A]; R05.Thm2 eroded composition with state-dependent contracts [Version B]; R05.Cor3 linear gain feasibility and small-gain condition; R05.Ex4 coupling-collapse counterexample; R05.Open5 the precise open problem). Target: **T5** ("useful assume–guarantee or nonlinear small-gain theorem allowing state-dependent interface contracts without hiding shared controls or nonconvex implementation").

Field 2 — Verdict

Partially closed by new restricted theorems; the general nonlinear small-gain problem remains genuinely open (precise statement with all missing hypotheses in Field 16 / R05.Open5). Version A and Version B are proved completely; the docket's acceptance elements are delivered: a true gain operator (the normalized interface-tightening matrix Γ), a true fixed point (the maximal feasible erosion vector $r^* = A^{-1}b$, computable by a contraction iteration), a counterexample outside the hypotheses (the coupling-collapse witness), and no insertion of nonlinear functions into numerical matrices (gains are declared Lipschitz constants of the true interface maps, never matrix-ized dynamics).

Field 3 — Exact statement

Data

N modules. Module i: state $x_i \in \mathbb{R}^{n_i}$; closed safe set Q_i with two-sided tubular radius $\rho_i > 0$ and $C^{1,1}$ signed distance in the tube (the erosion-lemma geometry of corrected 02 Lemma 2, including normal correspondence between ∂Q_i^{-r} and ∂Q_i). Coupled dynamics:

$$\dot{x}_i \in F_i(x_i, z_i(x), d_i) + \Delta_i B, \quad z_i(x) = (z_{ij}(x_j))_{j\neq i}, \quad d_i \in D_i,$$

with interface maps z_{ij} defined and Lipschitz on the ρ_j-tube of Q_j; exogenous error budgets $\Delta_i \geq 0$; compact disturbance sets D_i. Fix nominal interface values $\hat{z}_{ij}$ and define the **contract amplitudes**

$$\delta_{ij}(r) := \sup\{ \|z_{ij}(x_j) - \hat{z}_{ij}\| : x_j \in Q_j^{-r} \}, \quad 0 \leq r < \rho_j$$

(the guarantee module j offers at erosion depth r). Let $G_i(x_i) := \mathrm{clco}\{F_i(x_i, \hat{z}_i, d_i) : d_i \in D_i\}$ be the nominal envelope.

Hypotheses.

- **(H1)** Each G_i has nonempty compact convex values, is locally Hausdorff-Lipschitz near Q_i with constant L_i, and has linear growth (forward completeness).
- **(H2)** Boundary margin: for every $x_i \in \partial Q_i$ and $\zeta \in N^P_{Q_i}(x_i)$: $\sup_{v \in G_i(x_i)} \langle \zeta, v\rangle \leq -\alpha_i\|\zeta\|$ with $\alpha_i > 0$.
- **(H3)** Interface sensitivity: $\|F_i(x_i, z, d) - F_i(x_i, z', d)\| \leq \Lambda_i\|z - z'\|$ for all $d \in D_i$, uniformly on the tube of Q_i.
- **(H4)** The joint closed-loop envelope $G(x) := \mathrm{clco}\{ (v_i)_i : v_i \in F_i(x_i, z_i(x), d_i) \}$ satisfies the strong-invariance regularity (compact convex values, local Hausdorff-Lipschitz on a neighborhood of $\Pi_i\, Q_i$, growth) — automatic if F_i is jointly Lipschitz in (x_i, z) and each z_{ij} is Lipschitz.

R05.Thm1 (Version A — contract-amplitude composition)

If for every i

$$\Lambda_i \Sigma_{j\neq i} \delta_{ij}(0) + \Delta_i \leq \alpha_i,$$

then $Q := \Pi_i Q_i$ is strongly invariant for the coupled system: every solution starting in Q remains in Q for all forward time, under every disturbance realization. Each module's boundary margin absorbs the worst-case in-contract interface variation of its partners.

R05.Thm2 (Version B — eroded composition with state-dependent contracts)

If there exists an erosion vector $r \in \Pi_i [0, \rho_i)$ such that for every i

$$L_i r_i + \Lambda_i \Sigma_{j\neq i} \delta_{ij}(r_j) + \Delta_i \leq \alpha_i,$$

then $Q^r := \Pi_i Q_i^{-r_i}$ is strongly invariant for the coupled system. The condition is a genuine assume–guarantee loop: module i's admissible erosion r_i both consumes its own margin (term $L_i r_i$) and tightens the contracts it can guarantee its partners ($\delta_{ij}(r_j)$ decreasing in r_j — state-dependent interface contracts).

R05.Cor3 (linear gain feasibility — the small-gain condition)

Assume the **linear tightening bounds** $\delta_{ij}(r) \leq \delta_{ij}(0) - \sigma_{ij} r$ with $\sigma_{ij} \geq 0$ (contract tightens at rate σ_{ij} as module j retreats into its safe set). Define

$$b_i := \alpha_i - \Delta_i - \Lambda_i \Sigma_{j\neq i} \delta_{ij}(0) \quad \text{(base budgets)},$$
$$A := \mathrm{diag}(L_i) - M, \quad M_{ij} := \Lambda_i \sigma_{ij} \ (i \neq j), \quad M_{ii} := 0,$$
$$\Gamma := \mathrm{diag}(L_i)^{-1} M \quad \text{(the gain operator)}.$$

Then:

1. **(Fixed point / feasibility.)** If $\rho(\Gamma) < 1$ (spectral radius — the small-gain condition), then $A^{-1} = \Sigma_{k\geq 0} \Gamma^k \mathrm{diag}(L)^{-1} \geq 0$, the erosion fixed-point iteration $r^{(k+1)} := \mathrm{diag}(L)^{-1}(b + M r^{(k)})$ converges monotonically upward to $r^* := A^{-1}b$ from $r^{(0)} = 0$, and the feasibility condition of Thm2 (without the tubular cap) holds **iff** $A^{-1}b \geq 0$; r^* is the componentwise-maximal feasible erosion vector.
2. **(Certification.)** The composition is certified by Thm2 whenever $\rho(\Gamma) < 1$, $A^{-1}b \geq 0$, and $A^{-1}b < \rho$ componentwise (take $r = r^*$). If $A^{-1}b \geq \rho$ in some component, certification requires solving the capped feasibility (a finite linear program); if $A^{-1}b \ngeq 0$, no feasible erosion exists and the composition is refused.
3. **(Two modules, explicit.)** For $N = 2$: $\rho(\Gamma) < 1 \Leftrightarrow \Lambda_1\sigma_{12} \cdot \Lambda_2\sigma_{21} < L_1 L_2$ (the classical small-gain product), and $A^{-1}b \geq 0 \Leftrightarrow$

$$L_2 b_1 + \Lambda_1 \sigma_{12} b_2 \geq 0 \quad \text{and} \quad \Lambda_2 \sigma_{21} b_1 + L_1 b_2 \geq 0$$

(budget tradeoff: one module's deficit must be coverable by the partner's slack through the gain path). Version A is the special case $b \geq 0$ (no erosion needed). If both budgets are negative, no $r \geq 0$ is feasible — the loop cannot be closed at any depth.

R05.Ex4 (coupling-collapse counterexample — outside the hypotheses)

The positive linear system $\dot{x}_1 = -a_1x_1 + c_1x_2$, $\dot{x}_2 = -a_2x_2 + c_2x_1$, $x \geq 0$, with boxes $Q_i = [0, k_i]$ and $c_1c_2 > a_1a_2$: each isolated module ($c_i = 0$) has the full box as its viability kernel (decay to 0), but the coupled kernel of $Q_1 \times Q_2$ is exactly $\{(0,0)\}$ — every other nonnegative initial condition grows along the positive Perron eigenvector and exits any bounded box in finite time. The gain product $c_1c_2 > a_1a_2$ is precisely the violation of Cor3's small-gain condition (with $z_{ij} = x_j$, $\sigma_{ij} = 1$, $\Lambda_i = c_i$, $L_i = a_i$, $\delta_{ij}(r) = k_j/2 - r$), and the theorem correctly refuses certification at every erosion depth (the required depths exceed the tubular radii). Factorwise safety is destroyed by the interface loop — composition without a gain condition is unsound.

Field 4 — State and phase space

Product phase space $\Pi_i R^{n_i}$ (finite-dimensional ODE/differential-inclusion class); safe sets closed with tubular geometry; no delay, no events, no information state (deterministic full-state strong invariance — the weakest and cleanest class for a first composition theorem beyond corrected 03).

Field 5 — Quantifier order and information pattern

All-solutions strong invariance: $\forall x(0) \in Q^r\ \forall d(\cdot)\ \forall \text{solutions}: x(t) \in Q^r\ \forall t \geq 0$. One implicit feedback per module is allowed inside F_i (folded into the envelope); no disturbance-observing structure is assumed or needed. The assume–guarantee quantifier structure is static (contract bounds), not strategic: module j's guarantee is a set bound on z_{ij}, never a policy.

Field 6 — Assumptions, including existence/completeness

(H1)–(H4) as in Field 3; solutions of the joint inclusion exist (Filippov under (H4)) and are forward complete (growth in H1). The tubular/$C^{1,1}$ geometry excludes arbitrary closed sets — the packet's $\bigcup_j [2j, 2j+1]$ counterexample (corrected 02) remains controlling: without tubular regularity there is no uniform erosion calculus at all, hence no Version B.

Field 7 — Mapping type

EXACT_SPECIALIZATION of the erosion lemma (corrected 02) to product geometry, composed with the proximal-normal product calculus of corrected 03 — a genuine extension of the packet's composition toolkit from joint shared-control feasibility (corrected 03) to interface-contract composition; Ex4 is COUNTEREXAMPLE_OR_LIMIT.

Field 8 — Self-contained proof

Proof of R05.Thm1

Let $x \in \partial Q$ and $\zeta \in N^P_Q(x)$. For finite products, $N^P_Q(x) = \Pi_i N^P_{Q_i}(x_i)$ (the product formula already used by corrected 03 §2's proof). If $x_j \in \text{int } Q_j$ then $N^P_{Q_j}(x_j) = \{0\}$ (a ball around an interior point lies in Q_j), so ζ has nonzero components only on the active set $A(x) = \{i : x_i \in \partial Q_i\} \neq \varnothing$.

For $i \in A(x)$ and a realized velocity $v_i \in F_i(x_i, z_i(x), d_i) + \Delta_i B$: since $x_j \in Q_j$ for every j (as $x \in Q$), $\|z_{ij}(x_j) - \hat{z}_{ij}\| \leq \delta_{ij}(0)$, hence $\|z_i(x) - \hat{z}_i\| \leq \Sigma_{j\neq i} \delta_{ij}(0)$ (declared norm convention; any equivalent norm changes only constants). By (H3) there is $w_i \in F_i(x_i, \hat{z}_i, d_i)$ with $\|v_i - w_i - e_i\| = 0$ for some $\|e_i\| \leq \Delta_i$, so $v_i \in G_i(x_i) + (\Lambda_i \Sigma_{j\neq i} \delta_{ij}(0) + \Delta_i)B$. Using (H2) (homogeneous form on the proximal-normal cone):

$$\langle \zeta_i, v_i\rangle \leq \sup_{G_i(x_i)}\langle \zeta_i, \cdot\rangle + (\Lambda_i \Sigma_{j\neq i} \delta_{ij}(0) + \Delta_i)\|\zeta_i\| \leq (-\alpha_i + \Lambda_i \Sigma_j \delta_{ij}(0) + \Delta_i)\|\zeta_i\| \leq 0.$$

Inactive blocks contribute $\langle \zeta_j, v_j\rangle = 0$. Therefore $\sup_{v \in G(x)} \langle \zeta, v\rangle = \Sigma_{i\in A} \sup \langle \zeta_i, \cdot\rangle \leq 0$ (the inequality is preserved by closure and convexification). The proximal-normal strong-invariance criterion with (H1)/(H4) regularity yields strong invariance of Q (the same lemma corrected 02 Theorem 1 and corrected 03 §2 invoke). □

Proof of R05.Thm2

Let $x \in \partial Q^r$, $\zeta \in N^P_{Q^r}(x) = \Pi_i N^P_{Q_i^{-r_i}}(x_i)$, active set $A(x) = \{i : x_i \in \partial Q_i^{-r_i}\}$. For $i \in A(x)$: let $p_i \in \partial Q_i$ be the corresponding boundary point with common unit normal n_i (the tubular normal correspondence, corrected 02 Lemma 2 hypothesis), so $\|x_i - p_i\| = r_i$ and ζ_i is a nonnegative multiple of n_i on the regular part (and a proximal limit thereof generally — the inequality argument below is applied to proximal normals directly, exactly as in corrected 02 Lemma 2's proof). For a realized velocity $v_i \in F_i(x_i, z_i(x), d_i) + \Delta_i B$:

- since $x_j \in Q_j^{-r_j}$ for all j (as $x \in Q^r$), $\|z_{ij}(x_j) - \hat{z}_{ij}\| \leq \delta_{ij}(r_j)$;

- by (H3) and the envelope definition, $v_i \in G_i(x_i) + (\Lambda_i \Sigma_{j\neq i} \delta_{ij}(r_j) + \Delta_i)B$;
- by (H1)'s Lipschitz property, $\sup_{G_i(x_i)}\langle n_i, \cdot\rangle \le \sup_{G_i(p_i)}\langle n_i, \cdot\rangle + L_i\|x_i - p_i\| \le -\alpha_i + L_i r_i$ (the erosion computation of corrected 02 Lemma 2, clause-level match).

Combining:

$$\langle \zeta_i, v_i\rangle \le \|\zeta_i\| (-\alpha_i + L_i r_i + \Lambda_i \Sigma_{j\neq i} \delta_{ij}(r_j) + \Delta_i) \le 0$$

by the feasibility hypothesis of Thm2. Inactive blocks contribute zero. Convexification preserves the inequality; the strong-invariance criterion gives strong invariance of Q^r. □

Proof of R05.Cor3

Part 1. $\Gamma \ge 0$ entrywise. $\rho(\Gamma) < 1$ implies the Neumann series $\Sigma_{k\ge0}\Gamma^k$ converges, is entrywise ≥ 0, and equals $(I - \Gamma)^{-1}$. Since $A = \mathrm{diag}(L)(I - \Gamma)$:

$$A^{-1} = (I - \Gamma)^{-1} \mathrm{diag}(L)^{-1} = \Sigma_{k\ge0} \Gamma^k \mathrm{diag}(L)^{-1} \ge 0.$$

The fixed-point map is $\Phi(r) := \mathrm{diag}(L)^{-1}(b + Mr) = \Gamma r + \mathrm{diag}(L)^{-1}b$. From $r^{(0)} = 0$: $\Phi^k(0) = \Sigma_{m<k} \Gamma^m \mathrm{diag}(L)^{-1}b$, which is entrywise nondecreasing in k (each summand $\Gamma^m \mathrm{diag}(L)^{-1}b \ge 0$) and converges to $\Sigma_{m\ge0}\Gamma^m \mathrm{diag}(L)^{-1}b = A^{-1}b = r^*$. (For general b with negative components the iterates still converge to r^* by the contraction property of Φ; monotonicity holds whenever $\mathrm{diag}(L)^{-1}b \ge 0$.)

Feasibility equivalence (uncapped). If $r \ge 0$ and $Ar \le b$, multiplying by $A^{-1} \ge 0$ (order-preserving since entrywise nonnegative) gives $r \le A^{-1}b$, hence $A^{-1}b \ge 0$. Conversely, if $A^{-1}b \ge 0$, take $r^* := A^{-1}b$: then $Ar^* = b \le b$, so r^* is feasible. Hence $\{r \ge 0 : Ar \le b\} \neq \varnothing \Leftrightarrow A^{-1}b \ge 0$, and every feasible r satisfies $r \le r^*$: r^* is the componentwise-maximal feasible erosion vector.

Part 2. Immediate from Part 1 with the cap $r^* < \rho$ (any feasible $r \le r^*$ can be used if it fits; if r^* exceeds the cap, feasibility of the capped system is a finite linear program — exact but not closed-form).

Part 3. $N = 2$: $A = [[L_1, -\Lambda_1\sigma_{12}], [-\Lambda_2\sigma_{21}, L_2]]$; $\rho(\Gamma) < 1$ with $\Gamma = [[0, \Lambda_1\sigma_{12}/L_1],[\Lambda_2\sigma_{21}/L_2, 0]]$ gives $\rho(\Gamma)^2 = (\Lambda_1\sigma_{12}\Lambda_2\sigma_{21})/(L_1L_2) < 1$, the stated product. $\det A = L_1L_2 - \Lambda_1\sigma_{12}\Lambda_2\sigma_{21} > 0$, so $A^{-1} = (1/\det A)[[L_2, \Lambda_1\sigma_{12}],[\Lambda_2\sigma_{21}, L_1]] \ge 0$ and $A^{-1}b \ge 0$ expands to the two stated tradeoff inequalities. If $b_1 < 0$ and $b_2 < 0$ then $L_2b_1 + \Lambda_1\sigma_{12}b_2 < 0$ and $\Lambda_2\sigma_{21}b_1 + L_1b_2 < 0$: both fail — no feasible $r \ge 0$ exists (directly: at $r = 0$ both constraints are violated, and each constraint can only be repaired by the other variable, which is itself blocked). □

Verification of R05.Ex4

Isolated modules: $\dot{x}_i = -a_ix_i$ decays inside $[0,k_i]$ — full box viable. Coupled with $c_1c_2 > a_1a_2$: the Metzler matrix has a real eigenvalue $\lambda_+ > 0$ with positive right and left eigenvectors $v, w > 0$; for every $x_0 \ge 0, x_0 \neq 0$, $\langle w, x_0\rangle > 0$ and $x(t) \ge e^{\lambda_+ t}\langle w,x_0\rangle v/\|v\|\cdot v - (\text{lower modes})$ grows unboundedly along v, hence exits the bounded box in finite time; $(0,0)$ alone is an equilibrium. Kernel of the box under the coupled (uncontrolled) dynamics: $\{0\}$. Certificate side: $z_{ij}(x_j) = x_j$ gives $\delta_{ij}(r) = k_j/2 - r$ (nominal $\hat{z}_{ij} = k_j/2$), $\sigma_{ij} = 1$, $\Lambda_i = c_i$, $L_i = a_i$, $\alpha_i = a_ik_i - c_ik_j/2$ (requires the nominal to be contracting — if even the nominal escapes, $b_i < -\alpha_i$-type failure is immediate), $b_i = a_ik_i - c_ik_j$. $c_1c_2 > a_1a_2$ violates the product condition; in the symmetric case $a_i = a, c_i = c > a, k_i = k$ the feasibility inequality $(a-c)r \le (a-c)k$ requires $r \ge k > \rho = k/2$: refused at every admissible depth, matching the true collapse. □

Field 9 — Counterexample showing necessity or failure outside scope

Ex4 (gain loop) and the packet's shared-control infeasibility witness (corrected `03` §1: `u ≥ 1/2` vs `u ≤ −1/2` with empty intersection at the origin) are the two boundary witnesses — one for interface-gain failure, one for shared-control failure; both must be excluded by hypothesis, and they fail for different reasons (gain product vs. joint feasibility), which is why `TCS-1.0` §5's composition gate needs both theorems (CIRC-2). Outside scope additionally: nonconvex implementation (envelope convexification is a relaxation — realized trajectories are envelope solutions, but relaxation exactness is not claimed, per corrected `03` §7); non-tubular safe sets (no uniform erosion calculus — corrected `02`'s `⋃_j[2j,2j+1]` witness); delay/events (no claims).

Field 10 — Interface producer/consumer contract

- **Producer:** the assume–guarantee certificate: per module `(α_i, L_i, Λ_i, Δ_i, ρ_i)`, per interface `(δ_ij(·), σ_ij)`, plus the computed `(r*, Γ, ρ(Γ))` and the verdict.
- **Consumers:** R06 (scale composition: moment-closure approximations enter as `Δ`-budgets); the A023 spatial branch and polycentric composition docket (each spatial/jurisdictional interface must export `(z_ij, δ_ij, σ_ij)` records); Paper 2 (theorem + counterexample); the monograph composition chapter.
- **Failure condition:** certificate revoked if any exported constant is invalidated (e.g., a module's `α_i` recomputed after a model-version change — version identity axiom 7); using `Γ` entries as dynamics rather than Lipschitz constants violates the docket's "no nonlinear functions inserted into a numerical matrix without reduction" and is reviewer-rejectable.

Field 11 — Error, horizon, and safety erosion for approximations

The erosion triple is intrinsic: error budgets `Δ_i` (exogenous) and `Λ_iδ_ij(r_j)` (interface), erosion depths `r_i`, margins `α_i`, horizons infinite (strong invariance), tubular caps `ρ_i`. Approximate contracts (interface bounds known only up to `ε`) add `ε` to every `δ_ij(0)` — i.e. to the `b` vector — shrinking `r*` accordingly; the feasibility condition converts contract uncertainty into depth, exactly the erosion discipline the schema's GAP-3 proposes to type.

Field 12 — Selector and implementation regularity

No selector is needed: strong invariance is a statement about all trajectories of the coupled inclusion; the per-module feedback (if any) is inside `F_i` and its regularity is carried by (H1)/(H4) — the corrected `02` caution (measurable selection is not Lipschitz implementation) applies unchanged: a module claiming an implemented feedback must verify the envelope contains it.

Field 13 — Stochastic/hybrid/RFDE qualifications

Deterministic ODE inclusion class only. Hybrid resets void the tubular argument across events (reset preservation would need a jump-margin analogue — open); RFDE histories replace `x_j` by `x_{j,t}` with `δ_ij` defined on history sets — the translated-history closure of corrected `08` would be required (not done here); stochastic versions are chance-level statements needing QF-2 support alignment (not claimed).

Field 14 — Novelty status with exact references

Internal: extends corrected `03` from shared-control joint feasibility to interface-contract composition with erosion — no packet record does this; the docket's acceptance items (gain operator, fixed point, counterexample) are all present. External: small-gain theorems with gain operators and spectral-radius conditions are classical in robust control and ISS theory, and tubular/erosion arguments are standard in strong-invariance theory; **the specific packaging — contract amplitudes as interface suprema over eroded sets,**

feasibility as a linear system whose M-matrix property is the small-gain condition, and the maximal feasible erosion vector as the fixed point — requires external literature matching that this packet cannot provide (self-containment report caveat). No bibliographic novelty claim is made; the verdict "restricted new theorem, classical skeleton" is the honest status.

Field 15 — Publication destination

Paper 2 (abstract composition section: Thm1, Thm2, Cor3 with proofs; Ex4 beside corrected `03`'s counterexamples); monograph composition chapter; conditional docket for the nonlinear extension.

Field 16 — Remaining obligations and revocation triggers

R05.Open5 (the precise open problem — nonlinear small-gain with nonconvex implementation and shared controls). Statement target: an assume–guarantee theorem for `N` modules with (a) nonlinear gain operators `Γ_ij(r)` (nonlinear tightening functions, not linearized `σ_ij`), (b) set-valued nonconvex implementation correspondences inside `F_i`, (c) genuinely shared control channels (corrected `03`'s joint-feasibility clause), (d) non-tubular or nonsmooth safe sets. Missing hypotheses that any proof must supply: (i) a nonlinear fixed-point/contraction condition replacing `ρ(Γ) < 1` (candidate: a gain-loop composition condition in the sense of nonlinear small-gain theory — the external literature check of Field 14 is prerequisite); (ii) a relaxation-exactness or trajectory-level substitute for envelope convexification under nonconvex `I`; (iii) an erosion calculus for the declared set geometry (or a proof that tubular geometry is necessary — the `⋃[2j,2j+1]` witness suggests it is); (iv) joint feasibility of shared channels integrated with the contract loop (merging corrected `03`'s `R(x) ≠ ∅` clause with the gain condition — no published-in-packet proof exists). Other obligations: external novelty audit; instantiation of the certificate constants on one application pair (candidate: A023 spatial interface). Revocation triggers: invalidation of any exported constant (Field 10); discovery of an external theorem covering Version B in full (demotes to classical).

Field 17 — Machine-readable dependency edges


```
{
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  "target": "T5",
  "depends_on": [
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    "corrected_theorems/03_restricted_composition.md (product proximal-normal
    calculus, shared-control counterexample, relaxation-exactness caveat)"
  ],
  "unblocks": ["R06 (Δ-budgets for approximate aggregation)", "A023 spatial
  branch", "polycentric composition docket", "TCS-1.1 composition-gate
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  "status": {"R05.Thm1": "proved", "R05.Thm2": "proved", "R05.Cor3": "proved",
  "R05.Ex4": "proved counterexample", "R05.Open5": "open (hypotheses
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  "mapping_type": "EXACT_SPECIALIZATION + COUNTEREXAMPLE_OR_LIMIT",
  "novelty": "restricted new theorem on classical skeleton; external check
  outstanding"
}
```

Result Record R06 — Docket T6: Cross-Scale Aggregation and the Necessity of Memory

Field 1 — Result ID and target docket item

R06 (R06.Lem1 fibre-separation lemma; R06.Thm2 finite-augmentation obstruction; R06.Thm3 moment non-closure; R06.Cor4 approximate-closure erosion conversion; R06.Ex5 two-patch witness). Target: **T6** ("prove conditions under which coarse variables are dynamically closed, approximately closed, or necessarily memory-bearing … exact theorem plus obstruction; static expectation identity alone is insufficient").

Field 2 — Verdict

Already proved at the scoped level (projectability criterion, fibre obstruction, static aggregation identity — packet bases) + genuinely new negative result (finite moment closure is impossible for quadratic field dynamics; exact coarse variables are necessarily memory-bearing in the infinite-dimensional fibre regime). The positive approximate cases carry explicit defects; the erosion conversion ties them to safety.

Field 3 — Exact statement

Setting. Full system $\dot{x} = F(x)$ on a declared phase space X (finite-dimensional or function space), aggregate map $P: X \to R^m$, $m < \dim$, smooth on finite-dimensional X (on function spaces, P a declared moment functional). "Exact autonomous closure of the aggregate" means: a single-valued vector field g on $P(X)$ such that for **every** solution $x(\cdot)$ of the full system, $p(t) = P(x(t))$ solves $\dot{p} = g(p)$.

R06.Lem1 (fibre-separation lemma — lifted projectability). An exact autonomous closure of an augmented aggregate $q = (P, A)$ (with $A: X \to R^k$ smooth) exists only if the projected derivative $dP_x F(x)$ is constant on the fibres of q. In particular, if $x_1, x_2 \in X$ satisfy $q(x_1) = q(x_2)$ but $dP_{x_1}F(x_1) \neq dP_{x_2}F(x_2)$, no single-valued closure of q exists. (For A absent this is the A002 projectability criterion, `thm:projectability`, `sources/A002_general_theory_source.txt` line 1949, accepted in corrected `09`; the lemma records the lifted form needed for augmentation arguments.)

R06.Thm2 (finite-augmentation obstruction). If the fibres of P are infinite-dimensional in the sense that for some fibre $P^{-1}(p_0)$ and every finite k there exist pairs (x_1, x_2) in the fibre with $dP_{x_1}F(x_1) \neq dP_{x_2}F(x_2)$ that no continuous $A: X \to R^k$ separates at equal q-value (e.g. because the set of distinct projected-derivative values along the fibre has cardinality exceeding the separation capacity of R^k-valued continuous maps on that fibre), then no finite-dimensional augmented exact autonomous closure exists: **any exact evolution law for the aggregate is necessarily memory-bearing (history-dependent) or nonautonomous.** The exact closure that always exists is the trivial one on the full state — i.e. exact aggregation requires carrying the full field/history.

R06.Thm3 (moment non-closure for quadratic field dynamics). Let the full state be a nonnegative bounded field $X(\sigma)$, $\sigma \in \Sigma$ (Σ a probability space), with dynamics

$$\partial_t X(\sigma) = X(\sigma)(\rho(\sigma) - X(\sigma))$$

(pointwise spatial logistic; no transport). Let $m_k := \int X^k \, d\mu$ be the moments. Then:

1. **(Exact static identity.)** $\dot{m}_1 = \rho\bar{m}_1 - m_1^2 - \mathrm{Var}(X)$ — the A002 coarse-graining identity (proved, `thm:coarse-graining`, line 2094): the mean's derivative depends on the variance.
2. **(Moment recursion.)** $\dot{m}_k = k(\rho\bar{m}_k - m_{k+1})$ for every $k \geq 1$ (for constant ρ): each moment's derivative involves the next moment.
3. **(Non-closure.)** For **no** finite K does the moment family $(m_1, \ldots, m_K)$ admit an exact autonomous closure: for every K there exist pairs of initial fields with identi-

cal moments up to order K and different moments of order $K+1$ (explicit Chebyshev-alternation construction in the proof), so Lem1 applies. Consequently no finite family of moment functionals closes exactly; moment-based coarse variables are necessarily memory-bearing (the exact closure is the full field), and closures by arbitrary non-moment functionals are governed by Thm2's fibre-richness check, not granted. Approximate closures (moment-closure heuristics, support saturation) carry explicit defects (Cor4).

R06.Cor4 (approximate closure → safety erosion). If a reduced aggregate law $\bar{g}$ tracks the true aggregate with defect $\|\bar{g}(p) - (d/dt)P(x)\| \le \varepsilon$ on a compact domain over horizon T, then the aggregate trajectory error is at most $(\varepsilon/L)(e^{LT} - 1)$ (Grönwall, L the reduced law's Lipschitz constant), and safety claims certified through the aggregate transfer only via erosion: K_{-r} with the R03.Cor5 budget $\Delta = (\varepsilon/L)(e^{LT}-1)$. The packet's $O(\kappa)$ support-saturation result (A002, accepted in corrected 09: uniform vector-field defect $O(\kappa)$, finite-horizon stock error $O(\kappa)$) is the standing instance: approximate reduction is sound **only** with the horizon-eroded certificate.

R06.Ex5 (two-patch witness — the docket's obstruction instantiation). Two patches, $\dot{x}_i = g_i(x_i) + d(x_j - x_i)$, aggregate $p = (x_1+x_2)/2$. On the diagonal $x_1 = x_2$ (identical patches $g_1 = g_2$): the diagonal is invariant and $\dot{p} = g(p)$ closes exactly. With $g_1 \neq g_2$ (heterogeneous): at (x_1, x_2) and (x_2, x_1) (same mean), the mean-derivatives differ by $(g_1(x_1) + g_2(x_2) - g_1(x_2) - g_2(x_1))/2$, nonzero off the coincidence set for generic g_i; the fibre obstruction applies: no autonomous mean closure. Same structural axioms, opposite closure behavior — the R09.M3 witness.

Field 4 — State and phase space

Lem1/Thm2: declared phase space X (the theorem is phase-space-agnostic; the interesting case is function space for spatial fields). Thm3: $L^\infty(\Sigma, R_{\ge 0})$ with weakly continuous moment functionals; dynamics pointwise (a PDE-free spatial scaffold — transport only adds unclosed flux terms, worsening the obstruction). Ex5: R^2.

Field 5 — Quantifier order and information pattern

Closure statements quantify over **all** solutions of the full system (the closure must reproduce every trajectory's aggregate, not an equilibrium family or a measure-class average): $\forall x(\cdot) \in Sol: P(x(\cdot))$ solves $\dot{p} = g(p)$. This is the strongest (and honest) reading; weaker readings (closure along a single solution, closure in distribution) are approximations in the sense of Cor4, not exact closures.

Field 6 — Assumptions, including existence/completeness

Lem1: smoothness of P, F; uniqueness of full solutions on the stated domain (the A002 projectability criterion's hypotheses, corrected 09: autonomous full/reduced systems, C^1 projection, unique full/reduced solutions). Thm2: the fibre-richness hypothesis stated in the theorem (an explicitly checkable cardinality/separation condition). Thm3: $X \in L^\infty$ nonnegative, ρ constant (or $\rho^-$for the identity), moments finite; solutions exist pointwise. Cor4: compact domain, Lipschitz $\bar{g}$, defect uniform. Ex5: smooth g_i, $d \ge 0$.

Field 7 — Mapping type

Lem1: `EXACT_SPECIALIZATION` (lifted form of the proved projectability criterion). Thm2/Thm3: `COUNTEREXAMPLE_OR_LIMIT` (obstruction theorems). Cor4: `APPROXIMATION` with error/horizon/erosion triple. Ex5: `COUNTEREXAMPLE_OR_LIMIT`.

Field 8 — Self-contained proof

Proof of R06.Lem1

Suppose $q = (P, A)$ admits an exact closure $\dot{q} = \hat{g}(q)$ and let x_1, x_2 lie in a common q-fibre: $q(x_1) = q(x_2)$. Let $x_i(\cdot)$ be the (unique) solutions from x_i. At $t = 0$, the closure forces

$$dP_{x_i}F(x_i) = (d/dt)|_{0} P(x_i(t)) = \bar{g}_P(q(x_i)) = \bar{g}_P(q(x_1)) = dP_{x_1}F(x_1),$$

contradicting $dP_{x_1}F(x_1) \neq dP_{x_2}F(x_2)$. (For A absent this is exactly the necessity direction of the A002 projectability theorem — differentiating semiconjugacy at time zero — clause-level match.) □

Proof of R06.Thm2

Let $q = (P, A): X \to R^{m+k}$ be any finite-dimensional augmentation with A continuous, and suppose q admits an exact autonomous closure. By Lem1, dPF must be constant on q-fibres. The fibre-richness hypothesis provides, in a common P-fibre, two points x_1, x_2 with distinct projected derivatives that A does not separate. For such a pair, $q(x_1) = q(x_2)$ (same P-value by fibre membership; same A-value by hypothesis) while $dP_{x_1}F(x_1) \neq dP_{x_2}F(x_2)$: Lem1 contradicts the existence of the closure. Hence no finite k works, and the only exact closures are: the full state (memory of everything — trivially exact since P is a function of the state), history-dependent laws (memory-bearing), or explicitly nonautonomous laws (time-forced). □

Proof of R06.Thm3

(1) $\dot{m}_1 = \int X(\rho-X)d\mu = \rho m_1 - m_2 = \rho m_1 - m_1^2 - \mathrm{Var}(X)$ (the A002 identity $E[X^2] = \bar{X}^2 + \mathrm{Var}(X)$, proved at thm:coarse-graining; clause-level match).

(2) $\dot{m}_k = k\int X^{k-1}\cdot X(\rho-X)d\mu = k(\rho m_k - m_{k+1})$ for ρ constant (pointwise multiplication and dominated convergence on the bounded field).

(3) Fix $K \geq 1$. It suffices to construct two nonnegative bounded fields on a common probability space with equal moments $1..K$ and different moment $K+1$.

Construction (Chebyshev alternation + Vandermonde null vector). On $[0,1]$, let q be the best uniform approximation of x^{K+1} by polynomials of degree $\leq K$ (Chebyshev): the error $g := x^{K+1} - q$ attains $\pm\|g\|_\infty$ with alternating signs at $K+2$ alternation points $x_0 < x_1 < \ldots < x_{K+1}$, and $\|g\|_\infty > 0$ (since x^{K+1} has degree $K+1$). The $(K+1) \times (K+2)$ Vandermonde-type matrix with rows $(x_i^j)_{i}$ for $j = 0,\ldots,K$ has rank $K+1$, and its one-dimensional null space is spanned by the vector λ with

$$\lambda_i := 1 / \prod_{j\neq i}(x_j - x_i),$$

whose entries alternate in sign ($\mathrm{sign}(\lambda_i) = (-1)^i$ for ordered nodes, since exactly i of the factors $x_j - x_i$ are negative; the identities $\Sigma_i \lambda_i x_i^j = 0$, $j \leq K$, are the coefficients of the partial-fraction expansion of $1/\prod(x - x_i)$ at infinity). Define the signed measure $\sigma := \Sigma_i \lambda_i \delta_{x_i}$. By construction $\int x^j d\sigma = 0$ for $j = 0, \ldots, K$ (vanishing total mass and first K moments), while

$$\int x^{K+1} d\sigma = \int g\, d\sigma + \int q\, d\sigma = \Sigma_i \lambda_i g(x_i) \neq 0,$$

because each product $\lambda_i g(x_i)$ has one and the same sign (alternating × alternating) and $\|g\|_\infty \Sigma_i|\lambda_i| > 0$; the term $\int q\, d\sigma$ vanishes since q is a degree-$\leq K$ polynomial.

Now take $\Sigma = [0,1]$ with Lebesgue measure. Let μ be the uniform measure on the alternation points ($\mu(\{x_i\}) = 1/(K+2)$) and $\nu := \mu + \varepsilon\sigma$ with $\varepsilon > 0$ small enough that $\nu(\{x_i\}) = 1/(K+2) + \varepsilon\lambda_i \in (0,1)$ — a probability measure, distinct from μ, with the same first K moments and a different $(K+1)$-th moment. Define the step fields X^A (value x_i on a set of Lebesgue measure $\mu(\{x_i\})$) and X^B (value x_i on a set of Lebesgue measure $\nu(\{x_i\})$), both nonnegative and bounded. Then

$\int(X^A)^j\,du = \int(X^B)^j\,du$ for $j = 1, \dots, K$, and $\int(X^A)^{K+1}\,du \neq \int(X^B)^{K+1}\,du$.

By part (2), $\dot m_K(X^B) = K(\rho m_K - m_{K+1}(X^B)) \neq \dot m_K(X^A)$ although the moment states $(m_1,\dots,m_K)$ coincide at $t = 0$. By Lem1 (with $P = (m_1,\dots,m_K)$), no single-valued autonomous law $\bar g(m_1,\dots,m_K)$ can reproduce both fields' mean-moment dynamics: no finite moment family closes exactly. □

Proof of R06.Cor4

Grönwall comparison of the true aggregate trajectory against the reduced law's solution with the same initial value gives the trajectory bound; the erosion conversion is R03.Cor5 with Δ set to the trajectory error (the aggregate certificate is verified on K_{-r} where r absorbs the trajectory deviation). The $O(\kappa)$ support-saturation instance carries over verbatim from corrected 09 (uniform field defect $O(\kappa) \Rightarrow$ finite-horizon $O(\kappa)$ stock error), now with the explicit statement that the safety consequence is only the eroded certificate, on the fixed horizon — no bifurcation, memory, or governance transfer (matching the packet's scope limits for that theorem). □

Verification of R06.Ex5

Identical patches: on $\{x_1 = x_2\}$, $\dot x_1 - \dot x_2 = 0$, so the diagonal is invariant; on it $\dot p = g(p)$ with $g = g_1 = g_2$: exact closure on the invariant manifold (and the initial-condition class $x_1(0) = x_2(0)$ keeps trajectories there). Heterogeneous: at (a, b) and (b, a) (same mean $(a+b)/2$):

$$\dot m|_{(a,b)} - \dot m|_{(b,a)} = [g_1(a) + g_2(b) - g_1(b) - g_2(a)]/2$$

(the dispersal terms cancel in the mean), which is nonzero whenever $(g_1 - g_2)(a) \neq (g_1 - g_2)(b)$ — generic off the diagonal for $g_1 \neq g_2$. Two same-mean states with different mean-derivatives: Lem1's obstruction, instantiated. □

Field 9 — Counterexample showing necessity or failure outside scope

Ex5 and the Thm3 construction are the witnesses. Scope limits (retained from the packet): the projectability criterion rules out only exact autonomous Markovian reductions on the specified projection — approximate, memory-bearing, nonautonomous, stochastic, and singular reductions are not rejected (corrected 09's scope note); Thm3 strengthens this for the moment family (no finite augmentation at all), but says nothing about closures using non-moment functionals of the field (e.g., sufficient statistics for special field classes — closed only if the fibre-richness hypothesis of Thm2 fails there, which is exactly what must be checked per class).

Field 10 — Interface producer/consumer contract

- **Producer:** closure verdicts (exact / approximate-with-triple / impossible) per aggregate map P, with the Lem1 check (dPF constant on P-fibres?) as the cheap first test.
- **Consumers:** A023 spatial branch (moment-closure attempts must now cite Thm3: any finite moment closure is heuristic, and safety claims must route through Cor4's erosion); R09.M3 (witness supply); the monograph cross-scale chapter; Paper 2 (the obstruction theorem beside the A002 projectability results).
- **Failure condition:** using a moment-closure heuristic as an exact reduced model in any downstream theorem transfer (`REJECTED_MAPPING` enforceable by Thm3).

Field 11 — Error, horizon, and safety erosion for approximations

Cor4 is the complete triple: error $(\varepsilon/L)(e^{LT}-1)$ (or $O(\kappa)$ for support saturation), horizon T explicit (exponential degradation — no uniform claim), safety erosion r via R03.Cor5. Exact closures (diagonal case, projectable cases) carry no error.

Field 12 — Selector and implementation regularity

Not applicable: closure statements are about vector fields and trajectories, not policies. (Any governance built on an approximate aggregate inherits Cor4's erosion and R02's certificate discipline through the Δ-budget.)

Field 13 — Stochastic/hybrid/RFDE qualifications

Thm3 is deterministic pointwise field dynamics; stochastic field laws (superprocess-type) have the same moment-hierarchy obstruction a fortiori (moment recursion gains noise terms involving higher moments); hybrid resets on fields add jump-moment terms — both worsen closure, none is claimed as proved here beyond the deterministic statement. RFDE aggregates: the aggregate of a history-valued system may close only with history-valued g — the "memory-bearing" verdict in its RFDE guise; not proved here (declared open, aligned with the packet's delayed-hybrid gap).

Field 14 — Novelty status with exact references

Internal: the projectability criterion and fibre obstruction are proved and accepted (corrected `09`); the lifted augmentation obstruction (Lem1/Thm2) and the moment non-closure theorem with explicit separating fields (Thm3) are new in the packet's record space. External: moment-closure non-existence for multiplicative/quadratic field dynamics is classical folklore in spatial ecology (the packet itself cites `bolker1997`, `murrell2004` for "closure problems", A002 line 69 and line 2154's scope note: "neither result here supplies closed dynamics for the variance or covariance") — **Thm3's contribution is the precise finite-augmentation impossibility statement with a self-contained separating-field construction; external literature matching is outstanding**; no bibliographic novelty claim beyond that.

Field 15 — Publication destination

Paper 2 (beside the A002 projectability/aggregation block: Lem1, Thm3, Cor4); A023/Paper 7 conditional branch (Ex5 + Thm3 as the honesty boundary for any spatial moment programme); monograph cross-scale chapter.

Field 16 — Remaining obligations and revocation triggers

Obligations: external check of Thm3 against the moment-closure literature; RFDE-aggregate memory statement (open, Field 13); realizable-closure research programme (the A002 research-programme row on distributional closures stands — now constrained by Thm3 to non-moment or approximate families). Revocation triggers: discovery of a finite-dimensional sufficient statistic for a quadratic-field class satisfying the fibre-richness hypothesis (would falsify Thm2's application — check the hypothesis first); any use of a moment closure as exact.

Field 17 — Machine-readable dependency edges


```
{
  "result_id": "R06",
  "target": "T6",
  "depends_on": [
    "corrected_theorems/09_A002_reduction_diagnostic_audit.md (projectability
    criterion, fibre obstruction, static aggregation identity, O(κ) reduction)",
    "R03.Cor5 (erosion conversion for Cor4)",
    "R05 (Δ-budget interface for approximate composition)"
  ],
  "unblocks": ["R09.M3", "A023 spatial branch honesty boundary", "monograph
  cross-scale chapter"],
```

"status": {"R06.Lem1": "proved (lifted form of proved criterion)", "R06.Thm2": "proved", "R06.Thm3": "proved", "R06.Cor4": "proved (assembly)", "R06.Ex5": "proved witness"},
"mapping_type": "EXACT_SPECIALIZATION + COUNTEREXAMPLE_OR_LIMIT + APPROXIMATION",
"novelty": "obstruction theorems internal-new on classical folklore; external check outstanding"
}

Result Record R07 — Docket T7: Intergenerational Continuation and Specification Change

Field 1 — Result ID and target docket item

`R07` (R07.Def1 specification path — the proposed `TCS-1.1` type; R07.Thm2 generation recursion; R07.Prop3 fixed-specification reduction; R07.Thm4 alternating-disjoint impossibility and the necessity of typed resets; R07.Thm5 nested-compact existence; R07.Cor6 monotone-obligation finite horizon). Target: **T7** ("formalize generation-indexed safe sets, changing authorities/thresholds, continuation kernels, obligations, and architecture transitions").

Field 2 — Verdict

Repairable → proved (set level). The packet holds A001 §14's three results (verified below); what was missing is the typed specification-path object, the generation-indexed recursion with reset/obligation translation, the sharp impossibility showing typed resets are necessary (not merely convenient) for specification change, and the compactness existence complement. All proved here. No universal ethical claim is made (docket acceptance rule).

Field 3 — Exact statement

R07.Def1 (specification path — proposed typed object)

A **specification path** is `(G, (Ω_g)_{g∈G}, (T_g)_{g∈G})` where: `G = {0, 1, 2, …}` is the generation index with decision epochs `t_g` (locally finite: `t_g → ∞`); `Ω_g` is the frozen specification in force on `[t_g, t_{g+1})`; and `T_g = (τ^z_g, τ^h_g)` are **typed transition maps** at the generation boundary `t_{g+1}`: `τ^z_g` translates the phase state between architectures (`x ↦ R_g(x)`, set-valued, possibly architecture-changing — an Operator II reset), and `τ^h_g` translates the cumulative obligation/harm block with a declared variant from `{accumulate, cap, forgive}` (schema audit GAP-5). The generation-safe sets are `K_g := K_{q(g), Ω_g}`; intergenerational viability requires `x(t) ∈ K_g` for `t ∈ [t_g, t_{g+1})` along the whole path, with boundaries stitched by `τ_g`.

R07.Thm2 (generation-indexed continuation recursion)

Under the Operator II data conventions (corrected `04` §1) applied per generation with transition maps `T_g`, define

```
W_G := K_G  (terminal generation: maintenance condition, possibly an Operator I kernel)
W_g := { x ∈ K_g : ∃a ∀d :  Tube_g(x, a, d) ⊆ K_g   and   Succ_g(x, a, d) ⊆ τ_g^{-1}(W_{g+1}) }
```

(`τ_g^{-1}(W_{g+1})` = the set of phase states whose every translated successor branch lies in `W_{g+1}`). Then `W_g` is **exactly** the set of states from which a causal generation policy maintains every generation's safe set through generation `G` — the finite-horizon intergenerational continuation kernel — and the infinite-generation set is `W_∞ := ⋂_G W^{(G)}` (intersected over terminal horizons `G`, characterized by R03.Lem4's machinery under compactness).

R07.Prop3 (fixed-specification reduction)

If `Ω_g = Ω`, `K_g = K`, and `τ_g = id` for all `g`, then intergenerational viability equals ordinary viability on the concatenated horizon (A001 Theorem 14.1, `sources/A001_topdown_source.txt` line 1754 — trivial and verified: the generation index is a relabeling of time). Specification change is therefore exactly the added content of T7.

R07.Thm4 (alternating-disjoint impossibility; typed resets are necessary)

Let `K_{2k} = A`, `K_{2k+1} = B` with `A, B` closed and **disjoint** (`A ∩ B = ∅`), and `τ_g = id` (no reset: the state evolves continuously across the boundary). Then **no** intergenerationally viable trajectory exists, for any policy and any dynamics, regardless of generation lengths: continuity forces `x(t_{g+1}) ∈ closure(K_g) ∩ K_{g+1} = K_g ∩ K_{g+1} = ∅`. Conversely, with a typed reset `τ^z_g` whose range meets `K_{g+1}` (an architecture-changing translation), the recursion of Thm2 can be nonempty: **specification change across disjoint regimes is possible only through typed transformation, never through continuous evolution.** The necessity direction is the content: continuous dynamics cannot cross disjoint specifications — Operator II resets are not an optional modeling convenience but the only mathematical route.

R07.Thm5 (nested-compact existence)

If the `K_g` are nested decreasing (`K_{g+1} ⊆ K_g`) compact nonempty with `⋂_g K_g ≠ ∅`, trajectories are confined to a compact enclosure, the per-generation predecessor sets are nonempty in the sense of Thm2 at every finite `G`, and the successor maps satisfy the compactness/Hausdorff-upper-semicontinuity hypotheses of R03.Lem4, then `W_∞ ≠ ∅`: an infinite-generation continuation policy exists. (Complement of A001 Theorem 14.2, which proves impossibility when `⋂_g K_g = ∅` under compact confinement — verified: the subsequential-limit argument at line 1758–1760 is correct.)

R07.Cor6 (monotone obligations bound the generation count)

If the harm block translates by `accumulate` with a strictly positive obligatory increment `≥ c > 0` per generation (i.e., every admissible generation policy incurs it), and the normative floor is `h ≤ H_max` with `h ≥ 0`, then no intergenerationally viable path exists beyond `⌊H_max/c⌋` generations. (The A001 Corollary 9.1 integrability mechanism, line ~1660s region, in generation-indexed form.)

Field 4 — State and phase space

Phase space per generation: the architecture's typed product `Z_{q(g)}` (physical, obligation/harm, mode blocks); the path structure lives in the disjoint union `⊔_g {g} × Z_{q(g)}` (Operator II's phase-state convention, corrected `04` §1); transitions may change `q` through `τ^z_g`. Cumulative block `h` carried through `τ^h_g`.

Field 5 — Quantifier order and information pattern

Thm2: `∃a ∀d` per generation (meta-action before disturbance, exactly corrected `04`'s order); the generation policy is causal in the observed phase state at each `t_g`. Thm4: universal impossibility (`∀ policies ∀ dynamics` — the impossibility is information-independent). Thm5: `∃π ∀d ∀φ` infinite-horizon. No intergenerational welfare aggregation is performed anywhere — the recursion is set-valued viability, not optimization (the docket's "no universal ethical claim").

Field 6 — Assumptions, including existence/completeness

Thm2: Operator II data assumptions (exact tubes/successors, nonempty solution branches, at most one transition per interval — here exactly one, at the generation boundary). Thm4: closed `K_g`, continuous flow (no reset). Thm5: compactness + confinement + Hausdorff-usc successors (R03.Lem4's hypotheses). Cor6: positivity of the increment under every admissible policy (a strong hypothesis — flagged: if any policy can avoid the increment, the bound fails).

Field 7 — Mapping type

`TRANSFORMATION` (the whole record is an Operator II instance); Prop3 is `EXACT_SPECIALIZATION` to ordinary viability; Thm4's necessity is `COUNTEREXAMPLE_OR_LIMIT`.

Field 8 — Self-contained proof

Proof of R07.Thm2

Backward induction over generations, structurally identical to corrected `04` §5 (the packet's proved exact-tube recursion) with three replacements: the stage index becomes the generation index; the terminal set `G` becomes the terminal generation's maintenance kernel `K_G` (or an Operator I kernel when post-path maintenance is required); and the successor clause is composed with the typed translation: an endpoint branch `x'` at `t_{g+1}` succeeds iff every translated state `τ^z_g(x')` (and translated harm `τ^h_g`) lies in `W_{g+1}` — this is precisely `Succ_g ⊆ τ_g^{-1}(W_{g+1})`.

Sufficiency: at generation `g`, play the witness meta-action; every branch stays in `K_g` (tube clause) and lands in `τ_g^{-1}(W_{g+1})` (successor clause); the induction hypothesis at `g+1` continues from every translated branch. Necessity: a viable policy's first action must satisfy both clauses for every declared disturbance branch (a tube violation is a safety violation during generation `g`; a successor outside `τ_g^{-1}(W_{g+1})` lands outside `W_{g+1}`, from which — by induction — no continuation exists). The infinite-generation statement follows from R03.Lem4 applied to the generation-indexed predecessor chain under the stated compactness. □

Proof of R07.Thm4

Let `x(·)` be any continuous trajectory with `x(t) ∈ K_g` for `t ∈ [t_g, t_{g+1})`. At the boundary: `x(t_{g+1}) = lim_{t ↑ t_{g+1}} x(t)` with `x(t) ∈ K_g`, and `K_g` closed gives `x(t_{g+1}) ∈ K_g`; the next generation's constraint requires `x(t_{g+1}) ∈ K_{g+1}`. With `K_g ∩ K_{g+1} = ∅` this is impossible — for every policy, every dynamic, every generation length. (Convention check: whether the endpoint belongs to the closing or opening generation, one of the two memberships is forced by closedness and continuity, and the intersection is empty either way.) Conversely, if `τ^z_g` is a reset with `R_g(x) ∩ K_{g+1} ≠ ∅` on a nonempty set of states, the successor clause of Thm2 can be met on that set: the disjointness obstruction vanishes at the cost of a typed jump — which is exactly the Operator II semantics (state translation with identity/liability/obligation/harm carried by declared maps, not by continuous evolution). □

Proof of R07.Thm5

The finite-horizon kernels `W^{(G)}` are decreasing in `G` (a longer obligation can only shrink the viable set: `W^{(G+1)}`'s successor clause requires reaching `W^{(G)}`, and monotonicity of predecessors in their target is corrected `04` Corollary 1's pattern) and compact (closed subsets of the compact enclosure, by the closedness of predecessor sets under closed successor values). Nonemptiness at every finite `G` is the hypothesis. The intersection `W_∞ = ⋂_G W^{(G)}` is nonempty by the finite-intersection property of compact sets. That `W_∞` supports an infinite-generation policy and coincides with the set of states viable at all finite generation horizons is R03.Lem4's fixed-point argument, applied verbatim to the generation-indexed predecessor operator (the witness-action compactness extraction is the same). □

Verification of the retained A001 results

- Theorem 14.1 (line 1754): trivial relabeling — verified.
- Theorem 14.2 (lines 1758–1760): the subsequential-limit argument (`x(t_{k_j}) → x̄ ∈ K`, closedness of each `V^{(m)}`, `x̄ ∈ ⋂_m V^{(m)} = ∅`, contradiction) — verified

correct as stated, with the compact-confinement hypothesis explicitly retained (the source's own remark on its necessity stands).

- Theorem 14.3 (discounting pulse, lines 1766–1784): the welfare comparison $C(1 - e^{-\rho T}) > c_v \Leftrightarrow \rho > -(1/T)\ln(1 - c_v/C)$ — verified correct; retained as the behavioral separation (a discounting objective can prefer exit), not integrated into the viability recursion (the two judgments stay typed apart, matching `TCS-1.0`'s noncompensation discipline).

Proof of R07.Cor6

By induction: $h(t_{g+1}) = \tau^h_g(h(t_g)) \geq h(t_g) + c \geq h(0) + (g+1)c$ under `accumulate` with mandatory increment; the floor $h \leq H_{max}$ with $h(0) \geq 0$ gives $(g+1)c \leq H_{max}$, so $g \leq H_{max}/c - 1$. Beyond that generation count no admissible translated harm block satisfies the normative floor. □

Field 9 — Counterexample showing necessity or failure outside scope

Thm4 is itself the impossibility witness (disjoint alternation). Failure outside scope: if the generation boundaries are endogenous (events inside intervals rather than fixed epochs), Thm2's fixed-review recursion does not apply — the packet's standing variable-event gap; if τ^h_g = `forgive` is chosen, Cor6's bound vanishes (forgiveness resets the count — a modeling choice with normative content, not a theorem); if compactness fails, Thm5's conclusion can fail (escape-to-infinity, the packet's own remark on Theorem 14.2).

Field 10 — Interface producer/consumer contract

- **Producer:** the specification-path type + generation recursion + transition-map semantics (τ^z, τ^h variants).
- **Consumers:** `TCS-1.1` migration (GAP-1: the type is ready); the monograph's intergenerational chapter; any application claiming regime change (e.g., A018-style institutional transitions: their reset maps must be declared as τ^z_g records with harm translation — reviewer-enforceable); R09 Part U/M (the trichotomy fixed/change/disjoint is a scope statement, not a normative claim).
- **Failure condition:** applications asserting smooth (reset-free) transition between disjoint regimes — Thm4 makes this refutable in one line; applications using `forgive` without declaring it.

Field 11 — Error, horizon, and safety erosion for approximations

Not an approximation result (exact set-valued recursion). Approximate generation tubes follow corrected `04` Corollary 3's outer-tube discipline (inner certificates); erosion enters only through per-generation $K_{g,-r}$ if margins are needed — the R03/R05 machinery applies per generation unchanged.

Field 12 — Selector and implementation regularity

Arbitrary selectors (axiom-of-choice level) as in corrected `04`; measurable/regular generation policies need a selection theorem (open obligation D2, shared with R02). Implementation of τ^z_g (the reset) is an institutional act — its own implementation correspondence must be declared and all-branches-checked per R02's pattern before any intergenerational institutional claim.

Field 13 — Stochastic/hybrid/RFDE qualifications

The recursion is deterministic fixed-epoch; stochastic generation lengths or chance-constrained continuation require QF-2's support alignment and a chance-version of

Thm2 (open); RFDE phase states translate through `τ^z_g` as history resets (the review-synchronised resettable-memory semantics of corrected `08` is the compatible precedent); interior events remain outside scope.

Field 14 — Novelty status with exact references

Internal: A001 §14 supplies Definitions 14.1 and Theorems 14.1–14.3; nothing in the packet indexes specifications by typed paths with obligation translation, proves the disjoint-alternation necessity of resets, or states the nested-compact existence complement. External: generation-indexed viability and regime-switching safe sets have relatives in hybrid-systems and multistage-control literatures; **the typed-translation necessity statement (Thm4) and the accumulate/cap/forgive obligation semantics are, to internal knowledge, new packaging; external check outstanding**; no bibliographic claim made.

Field 15 — Publication destination

Paper 1 (intergenerational structure section: Def1, Thm2, Thm4); Paper 2 (theorem atlas: proofs); monograph intergenerational chapter (with A001 §14's verified results integrated and the discounting separation kept behavioral).

Field 16 — Remaining obligations and revocation triggers

Obligations: `TCS-1.1` adoption of the specification-path type; endogenous-boundary (variable-event) extension — open; stochastic continuation — open; selector regularity (D2). Revocation triggers: withdrawal of the fixed-epoch convention without a variable-event theorem; any application presenting `τ^h = forgive` as consequence-free (the normative content must be declared).

Field 17 — Machine-readable dependency edges


```json
{
  "result_id": "R07",
  "target": "T7",
  "depends_on": [
    "corrected_theorems/04_operator_II_transformation_candidate.md (exact-tube
    recursion, data conventions, monotonicity corollaries)",
    "corrected_theorems/08_A002_sampled_hybrid_audit.md (review-synchronised
    reset semantics for history states)",
    "R03.Lem4 (compactness horizon closure for Thm5)",
    "A001 §14 (Definition 14.1 line 1752; Theorems 14.1–14.3 lines 1754–1784) —
    verified"
  ],
  "unblocks": ["TCS-1.1 specification_path type (GAP-1)", "intergenerational
  applications", "monograph chapter"],
  "status": {"R07.Thm2": "proved", "R07.Prop3": "proved (trivial reduction,
  verified)", "R07.Thm4": "proved", "R07.Thm5": "proved", "R07.Cor6": "proved"},
  "mapping_type": "TRANSFORMATION + COUNTEREXAMPLE_OR_LIMIT",
  "novelty": "typed packaging internal-new; external check outstanding"
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Result Record R08 — Docket T8: Robust–Epistemic–Chance Hierarchy Completion (Typed Maps and Converse Counterexamples)

Field 1 — Result ID and target docket item

`R08` (R08.Prop1 exact-observation typed correspondence; R08.Ex2 five converse counterexamples (a)–(e); R08.Prop3 typed-map necessity). Target: **T8** ("state exact implications among robust, epistemic, institutional, and chance viability under support, filtration, information-state, and implementation assumptions … typed mappings rather than literal inclusions across different phase spaces; counterexamples to converses").

Field 2 — Verdict

Classical but useful (the monotonicity propositions, already fixed in corrected `01`) + new completion: the typed correspondence between physical and information-state judgments under exact observation, and five converse counterexamples closing every non-implication the hierarchy lists informally. Nothing here is mathematically deep; everything here is load-bearing for citation discipline, which is precisely the docket's demand.

Field 3 — Exact statement

Setting. The hierarchy's common signature `Ξ = (Ω, A_q, H, K, P, W, S)` (corrected `01` §Common signature); epistemic judgments on information states `B` with compatible sets `Compat(B)`; chance judgments under a declared law with filtration.

R08.Prop1 (exact-observation typed correspondence)

Let `O` be injective on a domain `D ⊇ K`, the filter exact (`B(x-history) = {x-history}` up to the observation identification), and the policy classes aligned (every observation-based policy on exact singletons is a state-feedback policy and conversely). Then the singleton map `Λ: x ↦ {x}` is a **typed correspondence**:

`x ∈ RViab(Ξ)  ⇔  Λ(x) ∈ EViab(Ξ_B),   and   Λ(RViab(Ξ)) = EViab(Ξ_B) ∩ Singletons.`

The epistemic kernel restricted to singleton information states is the physical robust kernel — through the map `Λ`, not as a subset statement. Under the same hypotheses with an implementation correspondence `I`, the institutional judgment corresponds on pairs `(Λ(x), c)`.

R08.Ex2 (converse counterexamples — each closes a hierarchy non-implication)

(a) Controlled ⇏ robust. `ẋ = u + d`, `K = [−1,1]`, `u ∈ [−1,1]`, `d ∈ [−2,2]`: `0 ∈ CViab` (policy `u ≡ 0` with branch `d ≡ 0` stays at 0) but `0 ∉ RViab` (against `d ≡ 2`, `ẋ = u + 2 ≥ 1` always: exit within time 1).

(b) Chance-`p` ⇏ robust (support inside the robust class). Same system, law `P(d ≡ 0) = 0.9`, `P(d ≡ 2) = 0.1`: `0 ∈ PViab_{0.9}` under `u ≡ 0` (safe with probability 0.9) while `0 ∉ RViab` — chance safety at `p < 1` is strictly weaker than robust safety even when the law's support is inside `W`.

(c) Robust ⇏ chance-1 (support exceeding the robust class). `W = {0}` (robust class: no disturbance), law `P(d ≡ 2) = 1`: `0 ∈ RViab(W = {0})` trivially, but `PViab_1` fails (the trajectory exits under the law's disturbance) — corrected `01` Proposition 8's guard is necessary, and the schema's QF-2 typing is what prevents the silent error.

(d) Physical nonempty ⇏ epistemic nonempty (typed failure). The hidden-mode system (A001 Example 4.1, line 558; R02.Prop3's plant): each physical state `(0, ±1)` is individually robustly viable, while the reachable information state `B = {(0,+1), (0,−1)}` lies in no epistemic kernel (empty common safe command). The physical kernel and the epistemic kernel are both nonempty and empty respectively — no inclusion can hold without the typed map, because the objects live on different spaces.

(e) Implementation enlargement shrinks all-branches safety (Prop 7 reversal). `K = {0}`, `ẋ = u + d`, `u ∈ {0}`, `d ≡ 0`: under `I(u) = {u}` the institution is safe (stay at 0); under the larger `I′(u) = {u, u+1}` the all-branches judgment fails (the branch `u+1 = 1` exits immediately). Enlarging implementation destroys all-branches institutional safety while enlarging some-branch safety — the quantifier flip is the whole content, and the schema's QF-3 guard (mandatory branch-quantifier declaration) is what makes the reversal visible before publication.

R08.Prop3 (typed-map necessity)

Without an explicit correspondence map, cross-space statements ("the epistemic kernel contains the physical kernel", "more information enlarges viability") are **ill-typed**: there is no subset relation between subsets of `X` and subsets of the information-state hyperspace, and any apparent inclusion smuggles in a map whose hypotheses (injectivity, filter exactness, policy lifting, commuting updates — corrected `01` Proposition 6's conditions) must then be proved. Every hierarchy comparison must therefore cite either an aligned signature (same space) or a declared map.

Field 4 — State and phase space

(a)–(c): `X = R`. (d): `X = R × {±1}` with information states in the hyperspace of closed subsets. (e): `X = R` with implementation correspondence. Prop1: physical space `X` vs. information hyperspace `B` (closed subsets of `X`).

Field 5 — Quantifier order and information pattern

(a): `∃π ∃w ∃φ` vs `∃π ∀w ∀φ` — the controlled/robust gap is exactly the disturbance quantifier. (b): `∃π: P(safe) ≥ p` vs `∃π ∀w`. (c): support condition `supp(P) ⊆ W` vs `⊄`. (d): `∃π^{obs}` against compatible-state universality. (e): `∃π ∀u^real ∈ I` vs `∃π ∃u^real ∈ I`.

Field 6 — Assumptions, including existence/completeness

All witnesses use Lipschitz dynamics with unique solutions; no existence subtleties. Prop1: injectivity of `O` on `D ⊇ K`, exact filter, policy-class alignment (all three hypotheses are exactly what corrected `01` Proposition 6 and A002's observation-fibre results already require — the packet's own observation-fibre criterion `K = O^{-1}(O(K))`, `thm:observation` line 501, is the static shadow of the same condition).

Field 7 — Mapping type

Prop1: `EXACT_SPECIALIZATION`/typed correspondence (an admitted mapping in `TCS-1.0` §7 terms — the bridge that makes hierarchy comparisons well-typed). Ex2: `COUNTEREXAMPLE_OR_LIMIT`. Prop3: schema-discipline statement (feeds QF-2/QF-3 guards).

Field 8 — Self-contained proof

Proof of R08.Prop1

Under injectivity of `O` on `D` and the exact filter, the information state generated by a physical history is the singleton of the observation-identified state: `B_t = Λ(x_t)`. Policy-class alignment: a policy on singletons is a function of the (identified) state, i.e. a state feedback, and every state feedback factors through the singleton information state. Hence closed-loop solution sets correspond one-to-one: `Sol(π, w)` maps onto `Sol(π^Λ, w)` for the corresponding policies, with identical disturbance classes and horizons. The robust judgment $\exists\pi\ \forall w\ \forall\varphi$ on the physical side translates clause-by-clause into the epistemic judgment $\exists\pi^\Lambda\ \forall w\ \forall\varphi$ on singleton information states (the compatible-state universality is vacuous on singletons). Kernel correspondence follows. The institutional extension adds the correspondence $(\Lambda(x),\ c) \mapsto$ I-branches, unchanged in structure. □

Verification of R08.Ex2

(a) $u \equiv 0,\ d \equiv 0$: $x \equiv 0 \in K$ — controlled viability. Against $d \equiv 2$: $\dot{x} = u + 2 \in [1,\ 3]$, so $x(t) \geq t$ and x exits $\{|x| \leq 1\}$ by $t = 1$ for every policy — not robust. **(b)** Under $u \equiv 0$ the safety event $\{d \equiv 0\}$ has probability $0.9 \geq p$: chance-viable at $p = 0.9$; robustness fails by (a)'s branch. **(c)** With $W = \{0\}$ the robust judgment quantifies over a single trivial branch; the law puts mass 1 on $d \equiv 2 \notin W$: the pathwise safety event has probability 0. **(d)** At $z = 0$ with $K = \{z \geq 0\}$: state $(0,+1)$ is maintained by $u = +1$ ($\dot{z} = +1$), state $(0,-1)$ by $u = -1$; from $B = \{(0,+1),(0,-1)\}$ every command crashes one branch — no observation-based policy is robustly safe, while each singleton is (R02.Prop3's plant at $z = 0$; the packet's own Example 4.1 verbatim). **(e)** $I = \{u\}$: trajectory stays at 0 — safe. $I' \supset I$: the realized-action set at the single command is $\{0,\ 1\}$; the branch $u^{real} = 1$ gives $\dot{x} = 1$, exiting $K = \{0\}$ immediately — the all-branches judgment fails under the larger correspondence. □

Proof of R08.Prop3

A subset statement $A \subseteq B$ requires $A,\ B$ to be subsets of a common space. Physical kernels are subsets of X; epistemic kernels are subsets of the hyperspace B. The only well-typed comparisons are (i) same-space comparisons (the hierarchy's Propositions 1–5, aligned signatures) or (ii) comparisons through a declared map Λ (Prop1's pattern), whose hypotheses must be proved. Corrected `01`'s Proposition 6 (information refinement) already encodes this discipline for refinement maps ("Without the policy-lifting and state-comparison maps, 'more information enlarges viability' is not a well-typed inclusion statement") — Prop3 generalizes the requirement to every hierarchy comparison and is enforced by the schema-audit guards QF-2/QF-3. □

Field 9 — Counterexample showing necessity or failure outside scope

Ex2 is the counterexample set; each is minimal. Failure outside scope: Prop1's correspondence requires injectivity on $D \supseteq K$ — A002's safety-crossing fibres (`thm:observation` corollary, line 537–542 region: two admissible states, same observation, opposite safety) are the standing obstruction to extending the correspondence beyond injective observations, and the packet's criterion is exactly the boundary: exact observation-only certification is possible iff safety is constant on observation fibres.

Field 10 — Interface producer/consumer contract

- **Producer:** the typed correspondence Λ + the counterexample register.
- **Consumers:** corrected `01` (the hierarchy document gains its converse counterexample appendix and the typed-map clause); Paper 2 (hierarchy section finalized); R09 Part U.2/M.6 (witnesses); every application paper that compares judgments across spaces (must cite Λ or an aligned signature — reviewer-enforceable).
- **Failure condition:** any publication asserting a literal inclusion between physical and epistemic/institutional kernels without a declared map (ill-typed per Prop3).

Field 11 — Error, horizon, and safety erosion for approximations

Not an approximation record. (Approximate observation/injectivity failure routes to R02's conservative filter soundness and R03's erosion; the correspondence degrades to outer/inner certificates exactly as there.)

Field 12 — Selector and implementation regularity

Prop1's policy correspondence is at the class level (no selector claims); Ex(e) is the implementation-regularity witness: the quantifier declaration is mandatory before any monotonicity claim (QF-3).

Field 13 — Stochastic/hybrid/RFDE qualifications

Ex(b)/(c) are the chance-qualification witnesses: support alignment (QF-2) is the typing that separates Proposition 8's valid direction from its failure; no filtration subtleties beyond support (the witnesses use constant-in-time laws). RFDE/hybrid: the correspondence pattern extends with the history-space machinery (corrected `08`) — declared, not proved here.

Field 14 — Novelty status with exact references

Internal: corrected `01` states Propositions 1–8 and eight non-implications informally (as warnings); this record supplies the typed correspondence and the explicit minimal witnesses — the completion the docket demands. External: monotonicity calculus and quantifier-gap counterexamples are standard robust-control/game folklore; **no external novelty claim is made or needed** — the value is citation discipline, and the external check is only to ensure the counterexample register doesn't duplicate a standard textbook list verbatim (irrelevant to truth).

Field 15 — Publication destination

Paper 2 (hierarchy section appendix: Prop1 + Ex2); corrected `01` successor document (the hierarchy record updated with this completion); monograph.

Field 16 — Remaining obligations and revocation triggers

Obligations: fold Ex2 into the corrected hierarchy record's next version; enforce QF-2/QF-3 in the `TCS-1.1` migration. Revocation triggers: none for the witnesses (they are concrete systems); Prop1 is revoked only if the correspondence hypotheses are dropped in citation.

Field 17 — Machine-readable dependency edges


```
{
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  "target": "T8",
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    "corrected_theorems/06_A001_selected_operatorI_audit.md (§2–§3: observation
    empties kernel, common-action obstruction)",
    "A002 thm:observation (fibre criterion, injectivity boundary)",
    "R02.Prop3 (hidden-mode witness reuse)"
  ],
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  "status": {"R08.Prop1": "proved", "R08.Ex2(a)-(e)": "proved witnesses",
  "R08.Prop3": "proved (typing discipline)"},
  "mapping_type": "EXACT_SPECIALIZATION (typed correspondence) +
  COUNTEREXAMPLE_OR_LIMIT",
  "novelty": "completion packaging; no external novelty claim"
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\newpage

Result Record R09 — Docket T9: The General-Theory Boundary Theorem

Field 1 — Result ID and target docket item

`R09` (R09.Thm1 Part U — universal consequences; R09.Thm1 Part M — six independence results). Target: **T9** ("a theorem or formal proposition stating exactly which claims are universal consequences of the typed axioms and which remain model-class dependent").

Field 2 — Verdict

Proved (new as a stated theorem). Part U assembles the packet's proved conditional laws into the exact universal list; Part M proves six independence results — for each candidate "universal law" of sustainability discourse, two axiom-consistent instantiations with opposite truth values. Together they are the boundary theorem that licenses the phrase "general theory" without implying universal delay, substitution, aggregation, or causal-diagnostic law (the docket's acceptance text, verbatim).

Field 3 — Exact statement

Axiom base. The seven structural axioms of `TCS-1.0` §9 (typed consistency; physical closure honesty; causal admissibility; noncompensatory admissibility; status monotonicity; interface explicitness; version identity), with the frozen judgment/mapping vocabulary. An *instantiation* is any admitted architecture realization `A_q` with declared specification `Ω` and solution concept satisfying the axioms.

R09.Thm1 (boundary theorem)

Part U — universal consequences. Every instantiation satisfies:

- **U1 (conservation conditional on closure):** for every declared boundary, moiety vector `L` in the left kernel of every active internal-flow and jump matrix, and locally finite execution, the telescoping identity `L^Tx(t) − L^Tx(0) = ∫_0^t L^TBφ ds + Σ_{t_j≤t} L^TB^Jβ_j` holds; conservation outside declared closures is asserted of nothing.
- **U2 (viability conditional on sets/classes):** the monotonicity calculus — robust ⊆ controlled; fixed-policy ⊆ robust; action-set expansion enlarges; disturbance-class expansion shrinks; safe-set expansion enlarges; robust ⊆ chance-1 under support alignment — in every aligned signature.
- **U3 (noncompensation conditional on declared binding components):** componentwise failure of a binding constraint is not erased by unrelated surplus unless `Ω` declares an authorized substitution pathway; on the restricted linear domain, feasibility is exactly Farkas-certified, pathway-specifically.
- **U4 (status/interface discipline):** no integration step strengthens evidence status; no cross-module transfer without an admitted mapping + contract.
- **U5 (kernel-recursion exactness under stated hypotheses):** where the sampled/held/information-state hypotheses hold, the kernel recursions characterize their judgments exactly.

Part M — independence results. None of the following is a consequence of the axioms; each is refuted by a pair of axiom-consistent instantiations with opposite truth values:

- **M1 (no universal delay law).** Instantiation I: `ẋ = −2x(t) − x(t−τ)` — delay-independent exponential stability for every `τ ≥ 0` (Halanay certificate, A002 `thm:small-gain`, proved/accepted). Instantiation II: `ẋ = −x(t−τ)` — instability for every `τ > π/2` (explicit growing solution `e^{λt}` with `Re λ > 0`, constructed below). "Delay destabilizes" and "delay is harmless" both fail as axiom-level claims.
- **M2 (no universal substitution law).** Feasible pathway: single-resource linear endowment `x = 3 ≥ s^req = 2` (allocation `a = 2` satisfies all constraints). Infeasible pathway: `x = 1 < s^req = 2` with the strict Farkas separation `γ^Ts^req > α^Tx + β^Te` (A002 `thm:farkas`, proved). Substitution is pathway-specific in both directions; no exchange rate exists.
- **M3 (no universal dynamic aggregation).** Identical-patch system: the aggregate mean closes exactly on the invariant diagonal. Heterogeneous two-patch system (R06.Ex5): fibre obstruction — no autonomous mean closure; and R06.Thm3: no finite moment family closes for quadratic field dynamics. Static expectation identities are exact; dynamic closure is model-class-dependent.
- **M4 (no universal causal-diagnostic law).** Injective observation on the declared domain: exact safety certification (A002 `thm:observation` sufficiency). Safety-crossing fibres: no observation-only certificate exists (same theorem's obstruction side). Stock-to-rate margins: unboundedly unsound without rate-persistence (R03.Thm2). Diagnostics are certificates only under declared conditions; causal reading is never automatic.
- **M5 (no universal local-to-global law).** A fold occurring in an uncoupled coordinate leaves an independent safe set's kernel unchanged (explicit 2-D witness below). The A018-class systems exhibit folds interacting with constraints (source-stated status, not promoted). Local bifurcation existence neither implies nor is implied by global safe-set change.
- **M6 (no universal information/implementation monotonicity).** Refinement monotonicity holds only through typed lift maps (corrected `01` Prop 6; R08.Prop1); untyped "more information is better" fails (delayed revelation changes nothing at the decision point — the delayed-information obstruction, corrected `06` §4); implementation enlargement shrinks all-branches safety (R08.Ex(e)).

Conclusion. The universal content of the typed general theory is exactly the conditional laws U1–U5 plus the obstruction calculus; every predictive mechanism — delay effects, substitution, aggregation closure, diagnostic causality, bifurcation-safety links, information/implementation monotonicity — is model-class-dependent, with the independence witnesses above as the permanent record of that boundary.

Field 4 — State and phase space

Part U: phase-space-agnostic (the laws are typed over any declared class). Part M witnesses: M1 scalar RFDE `C([−τ,0], R)`; M2 finite-dimensional linear allocation space; M3 `R²` and field space `L^∞(Σ)`; M4 `R`/`R×{±1}` with observation maps; M5 `R²`; M6 as in R08.

Field 5 — Quantifier order and information pattern

Part U: each law's quantifier structure is the packet's own (U1: universal over executions; U2: the hierarchy's aligned `∃π ∀w ∀φ` chains; U3: universal over binding components; U4: universal over integration steps; U5: hypothesis-conditional equivalences). Part M: each independence result is a *pair* of `∃`-witnesses — the logical form is `¬(Axioms |= Claim)`, demonstrated by two models of the axioms disagreeing on the claim.

Field 6 — Assumptions, including existence/completeness

Part U: exactly the hypotheses of the assembled theorems (A002 conservation theorem's local finiteness and moiety conditions; hierarchy alignment; Farkas domain restrictions; sampled-kernel compactness/continuity). Part M: each witness is a well-posed Lipschitz system with unique solutions (M1: linear RFDE, classical well-posedness).

Field 7 — Mapping type

Part U: `EXACT_SPECIALIZATION` (assembly of proved laws into the boundary statement). Part M: `COUNTEREXAMPLE_OR_LIMIT` (independence witnesses). The theorem as a whole is the docket's `T9` deliverable and the `ANALOGY_ONLY` firewall: any model-class mechanism cited as universal is exactly one of M1–M6 and is refutable by the corresponding witness.

Field 8 — Self-contained proof

Part U

Each item is a packet theorem restated at the axiom level; the assembly adds only the observation that the laws' hypotheses are themselves typed declarations (closure, moiety, alignment, domain, registry), so the laws hold in *every* instantiation *as conditional statements*:

- U1: A002 `thm:conservation` (statement lines 227–241; proof 243–263; accepted with the `Bφ` notation correction in corrected `07`, which also verified the left-kernel condition across active modes). The identity is a trajectory-level algebraic identity — universal over all executions satisfying the declared structure.
- U2: corrected `01` Propositions 1–5 and 8 (each proved there; R08 adds the converse witnesses showing the implications are strict).
- U3: A002 domain-qualified noncompensation (statement lines 192–199; proof 201–220; accepted) + `thm:farkas` (statement 428–442; proof 444–483; accepted with the dimension check in corrected `07`).
- U4: axioms 5–6 of `TCS-1.0` §9 (definitional, enforced by the theorem-record schema).
- U5: corrected `08`'s accepted kernel chain, each at its exact restricted status.

No new mathematics; the theorem's content is that **this list is complete for the programme's universal claims** — which Part M establishes negatively, by refuting every candidate extension. □

Part M

M1. Instantiation I is the packet's proved Halanay result (corrected `09`'s adjudication row: `α_0 > β_0` gives delay-independent exponential decay — for the scalar system `ẋ = −2x(t) − x(t−τ)`, `α_0 = 2 > 1 = β_0` in the standard norm). Instantiation II: consider `ẋ(t) = −x(t−τ)` and the characteristic equation

$$\lambda + e^{-\lambda\tau} = 0.$$

At `τ = π/2`, `λ = i` is a root (`i + e^{−iπ/2} = i − i = 0`). The root is simple: `∂F/∂λ = 1 − τe^{−λτ} = 1 + τλ = 1 + iπ/2 ≠ 0` for `F(λ,τ) = λ + e^{−λτ}` (using `e^{−λτ} = −λ` along the root locus). By the implicit function theorem the root continues as `λ(τ)` near `π/2` with

$$d\lambda/d\tau = -(\partial F/\partial\tau)/(\partial F/\partial\lambda) = -(\lambda^2)/(1 + \tau\lambda),$$

(since $\partial F/\partial\tau = -\lambda e^{-\lambda\tau} = \lambda^2$). At $(i, \pi/2)$: $\lambda^2 = -1$, so $d\lambda/d\tau = 1/(1 + i\pi/2) = (1 - i\pi/2)/(1 + \pi^2/4)$, whose real part is $+1/(1+\pi^2/4) > 0$: the root crosses into the right half-plane as τ increases through $\pi/2$. Hence for every $\tau \in (\pi/2, \pi/2 + \delta)$ there is a root λ with $\mathrm{Re}\ \lambda > 0$, and indeed by continuity of the crossing this holds for a definite interval; for $\tau > \pi/2$ (up to the next return crossing, and in particular on an explicit interval) the zero solution is unstable. The corresponding exponential $x(t) = e^{\lambda t}$ (with the consistent history $\varphi(s) = e^{\lambda s}$, $s \in [-\tau, 0]$) is an exact solution of the RFDE growing without bound: an explicit instability witness (no stability theorem invoked — the growing solution *is* the proof). Both instantiations satisfy the axioms (typed single-stock RFDE, declared delay, closed history phase space); the truth values of "the delay destabilizes" are opposite. □

M2. The feasible instance: one resource, endowment $x = 3$, requirement $s^{req} = 2$, $R = E = I$, $Q = I$: $a = 2$ satisfies $Ra \le x$, $Qa \ge s^{req}$, $a \ge 0$ — substitution (service met through the pathway) is possible. The infeasible instance: $x = 1$, same requirement: $a \le 1$ and $a \ge 2$ — no feasible allocation; the Farkas alternative supplies the certificate $\gamma\cdot 2 > \alpha\cdot 1$ with $\gamma = \alpha = 1$ ($2 > 1$). Both are axiom-consistent linear instantiations with identical structure and different endowments: substitution feasibility is not a structural property. □

M3. R06.Ex5 (proved there): identical patches close the mean exactly on the invariant diagonal; heterogeneous patches obstruct any autonomous mean closure (same mean, different mean-derivatives). R06.Thm3 (proved there): for quadratic field dynamics no finite moment family closes. Static aggregation identities (A002 `thm:coarse-graining`, proved) hold in both instances — the static/dynamic boundary is exactly the independence boundary. □

M4. The A002 observation-fibre criterion (proved, accepted in corrected `07`) gives both directions: $K = O^{-1}(O(K))$ relative to the declared domain makes observation-only certification exact; a safety-crossing fibre (two admissible states, same observation, opposite safety membership) obstructs every observation-only certificate — the theorem's own corollary (lines 537–542) supplies the obstruction instance. R03.Thm2 (proved) adds the margin side: stock-to-rate diagnostics are unboundedly unsound without rate persistence. Two instantiations, opposite diagnostic validity — no causal-diagnostic law at axiom level. □

M5. Witness for "fold without global safe-set change": on $\mathbb{R}^2$,

$$\dot{x}_1 = \mu + x_1^2,\quad \dot{x}_2 = -x_2 + c,\quad c = 1/2,$$

with safe set $K = \{x_2 \le 1\}$ (no constraint on x_1). The x_1-block undergoes a saddle-node (fold) at $\mu = 0$ — a genuine local bifurcation of the declared model class. The x_2-block: from any $x_2(0) \le 1$, $x_2(t) = (x_2(0) - 1/2)e^{-t} + 1/2 \le 1$ for all $t \ge 0$ — every trajectory from K stays in K forever, independent of μ and of the fold: $\mathrm{Viab}(K) = K$ for every μ. The fold exists and the kernel never changes. The opposite direction (folds interacting with constraints, A018 C3/C4-class evidence) exists in the packet at source-stated status — not promoted, not needed for the independence claim: the pair "fold-without-kernel-change" (proved here) versus "no-fold-with-kernel-collapse" (e.g., $\dot{x} = 1$ on $K = [0,1]$: no bifurcation, kernel collapses to $\{1\}$… in fact empty of interior; either way trivially opposite) establishes that neither property implies the other at axiom level. □

M6. The typed-map necessity (R08.Prop3, proved) plus the delayed-information obstruction (corrected `06` §4: accurate information arriving after `T_obs > q_0/ε` cannot prevent exit — the obstruction is proved in the packet) and the implementation reversal (R08.Ex(e), proved). "More information/implementation is better" fails in both directions without the typing. □

Field 9 — Counterexample showing necessity or failure outside scope

Part M *is* the counterexample register — the theorem's method is independence witnessing. Scope note: Part U's completeness is relative to the programme's claim inventory (the master prompt's list of universal candidates); a new candidate universal claim must be either proved from the axioms (joining Part U) or added to Part M's witness list — the theorem provides the test, and the packet's status discipline (axiom 5) provides the enforcement.

Field 10 — Interface producer/consumer contract

- **Producer:** the boundary statement (U-list + M-register).
- **Consumers:** the flagship Paper 1 (the theory's scope section is a citation of this theorem); the monograph (the general-theory claim is bounded by it); every application paper (model-class mechanisms must cite their M-item, which converts "this is generally true" into "this is a model-class mechanism with witnesses"); the novelty audit (F1: the theorem delimits what needs external comparison — the conditional laws and the boundary, not the mechanisms).
- **Failure condition:** any downstream text citing a Part-M mechanism as universal, or a Part-U law without its conditional's typed hypotheses — both are reviewer-enforceable rejections.

Field 11 — Error, horizon, and safety erosion for approximations

Not an approximation result; the erosion discipline enters U-list items only through their own theorems (U5's kernel results and the erosion conversions of R03/R05 when instantiated).

Field 12 — Selector and implementation regularity

No selector claims; M6's implementation witness is itself the regularity discipline statement (QF-3).

Field 13 — Stochastic/hybrid/RFDE qualifications

M1's instability witness is an RFDE instantiation (axiom-consistency across model classes is part of the point: the boundary theorem does not privilege the ODE class); the Halanay certificate is likewise RFDE. No stochastic witnesses are needed for the six listed claims; a candidate stochastic universal law (e.g., "noise always destroys infinite-horizon viability") is already conditional in the packet (A001 Conjecture 17.2 remains conjectural, correctly) and would join Part M's register if promoted.

Field 14 — Novelty status with exact references

Internal: no packet record states the boundary as a theorem; the master prompt's task 8 and the docket T9 demand exactly this statement; the U-items are packet-proved, the M-witnesses are proved in R01/R03/R06/R08 or here (M1's crossing computation and M5's witness are new verifications; M1-I, M2, M3, M4, M6 reuse packet-internal proved results). External: scope/boundary theorems of this "conditional universality + independence witnesses" shape exist in axiomatic-modeling traditions; **the sustainability-specific witness register and the exact U-list are, to internal knowledge, new; external literature check outstanding**; no bibliographic claim made.

Field 15 — Publication destination

Paper 1 (the theory's central scope theorem — likely its most-cited statement); Paper 2 (the witness proofs); monograph (opening chapter: what the general theory claims and does not).

Field 16 — Remaining obligations and revocation triggers

Obligations: external novelty audit of the boundary-theorem form; maintenance discipline — every future candidate universal claim must be routed through the U/M test before publication; the M-register is append-only (new mechanisms join with witnesses). Revocation triggers: discovery that a Part-U law's assembled proof has a gap (would demote that item to conditional and shrink the universal list); discovery of an axiom-level proof of any Part-M claim (would falsify the corresponding witness — mathematically impossible for the proved witnesses, but the register's framing must track any axiom change in `TCS-1.1`).

Field 17 — Machine-readable dependency edges

```json
{
  "result_id": "R09",
  "target": "T9",
  "depends_on": [
    "A002 thm:conservation, thm:farkas, thm:observation, thm:coarse-graining,
    thm:small-gain (U1, U3, U4/M2/M4, M3, M1-I)",
    "corrected_theorems/01_operator_I_hierarchy.md (U2)",
    "corrected_theorems/07,08,09 (acceptance records for the U-items)",
    "corrected_theorems/06 §4 (M6 delayed-information)",
    "R01.Thm2 (M3 aggregate witness)", "R03.Thm2 (M4 margin witness)",
    "R06.Ex5+Thm3 (M3)", "R08 (M6, U2 converses)"
  ],
  "unblocks": ["Paper 1 scope section", "monograph general-theory claim",
  "novelty audit F1", "TCS-1.1 axiom maintenance discipline"],
  "status": {"R09.Thm1 Part U": "proved (assembly of proved laws)", "R09.Thm1
  Part M.1–M6": "proved (independence witnesses)"},
  "mapping_type": "EXACT_SPECIALIZATION + COUNTEREXAMPLE_OR_LIMIT",
  "novelty": "boundary-theorem packaging internal-new; external check
  outstanding"
}
```